# Theoretical Minimum
# Advanced Quantum Mechanics

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**Abstract**

These are my notes from the Advanced Quantum Mechanics lectures from Leonard Susskind's Theoretical Minimum series[1, Advanced Quantum Mechanics].

Disclaimer: I have prepared these notes as an aide-mémoire for my own use; if you find them useful, you are welcome, but I'd appreciate hearing from you. They are not intended as a substitute for listening to the lectures. The intellectual property for all material derived from the lectures belongs, of course, to Professor Susskind; any mistakes, however, are my own.

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# 1 Review and Introduction to Symmetry

We start with a review of quantum mechanics, and then move on to the applications–not the technology but the basic physics problems for which quantum mechanics was invented: atoms; electrons, which stand for a whole class of particles, muons, quarks, neutrinos; and photons. One theme will be symmetry–how symmetries are realized in quantum mechanics, how they are represented, and what they tell us about the quantum mechanical system

## 1.1 Review of Quantum Mechanics

This is a review of material from [1, Quantum Mechanics] and [2].

- We start by asking how to represent the state of a system: in classical mechanics states are represented as points in phase space, in quantum mechanics they are represented by state vectors, in a vector space–$|\psi\rangle$ and $\langle\psi|$. Complex numbers are ubiquitous in quantum mechanics, and our vector space is also complex; $\langle\psi|$ is analogous to the complex conjugate of $|\psi\rangle$.

- Observables. We have observables in classical mechanics, such as position and momentum, but we assume we know what we mean when we say "observable". In quantum mechanics we have to be very precise. An observable is represented by a Hermitian matrix. Hermitian operators are analogous to real numbers: $A = A^\dagger$.

- Eigenvalues represent values that can be measured: $A\,|\alpha\rangle = \alpha\,|\alpha\rangle$. The set of eigenvalues is the set of possible values of a measurement; the eigenvectors are the state vectors of the system such that, if you make a measurement in one of these states, the value is definite, not statistical, and the value is $\alpha$.

- Inner product $\langle\phi|\psi\rangle$– a complex number defined for any bra/ket pair.

- Orthogonal $\langle\phi|\psi\rangle = 0$–for any pair of states that can be distinguished by some possible measurement.

- A particle is something that has a definite position in space. The position, $x$ say, should be thought of as an observable. If we measure a particle's position and definitely find it at $x_0$, we have state vector $|x_0\rangle$.

- Independent values $\langle x|x'\rangle = 0$ if $x$ and $x'$ are distinguishable.
- For any state $|\psi\rangle$, $\langle x|\psi\rangle = \psi(x)$ is called the *wave function*.
- The meaning of $\psi$ is closely related to probability: probability of finding particle at $x$ is $P(x) = \psi^*(x)\psi(x)$
- Position is an observable $X$. In 3 dimensions $\langle xyz|x'y'z'\rangle = 0$ if there is any mismatch $x' \neq x \vee y' \neq y \vee z' \neq z$. We will generally suppress $y$ and $z$, but remember they are still there.
- Another important observable is Momentum, which has one component for each dimension of space: $P\psi(x) = -i\hbar\frac{\partial\Psi}{\partial x}$.
- The eigenvector of $X$ corresponding to $x_0$ is $\delta(x - x_0)$.
- The eigenvector of $P$ corresponding to $p_0$ is $e^{\frac{ip_0x}{\hbar}}$. Using $P(x) = \psi^*(x)\psi(x)$, $P(x) = 1$–particles is smeared along entire line–Uncertainty Principle.

- Evolution operator $U(t)$:
  - $U(t)\,|\psi(0)\rangle = |\psi(t)\rangle$
  - $U(t)\,|\psi(t_0)\rangle = |\psi(t_0 + y)\rangle$
- Following the "minus 1-th law"[1], we postulate that two states that are observably different will not evolve into two other states that are *not* observably different, so inner products must be preserved: $\langle\phi|\psi\rangle = \langle\phi|U^\dagger U|\psi\rangle$ i.e. $U^\dagger U = I$. U is *Unitary*
- Let $U(\epsilon) = I + \epsilon G$, so $U^\dagger(\epsilon) = I - \epsilon G^\dagger$ and $G = -G^\dagger$. This, in turn, requires $(I + \epsilon G)(I - \epsilon G^\dagger) = I + O(\epsilon^2)$. We can satisfy this be defining $G = -iH$ for some Hermitian $H$–the *Hamiltonian*.
- For general $t$, $U(t) = e^{-iHt}$

$$\begin{aligned}
\psi(t+\epsilon) =& e^{-i\hbar\epsilon}\,|\psi(t)\rangle \\
=& (1 - i\hbar\epsilon)\,|\psi(t)\rangle \\
\frac{|\psi(t+\epsilon)\rangle - |\psi(t)\rangle}{\epsilon} =& -i\hbar\,|\psi(t)\rangle \\
\frac{\partial\,|\psi\rangle}{\partial t} =& -iH\,|\psi(t)\rangle\text{, Time dependent Schrödinger Equation} \\
H\,|\psi\rangle =& E\,|\psi\rangle\text{, Time independent Schrödinger Equation}
\end{aligned}$$

How does eigenvector change with time? It just gets multiplied by a *phase*, which has no effect on probability.

[1] Information is conserved

## 1.2 Introduction to Symmetry

Evolution is one example of a transformation on a system which preserve inner products.

- One important example is the rotational symmetry of the hydrogen atom.
- Translational symmetry is another example.

**Definition 1** (Symmetry)**.** *A symmetry is a transformation that doesn't change equations.*

If $V$ is a symmetry we expect it to be unitary, as it should preserve differences between wave functions.

**Theorem 2** (A symmetry is a unitary operator that commutes with the Hamiltonian)**.**

$$V|\psi\rangle = |\psi'\rangle \textit{ is a symmetry} \tag{1}$$

$$\iff$$

$$VH = HV \tag{2}$$

*Proof.* Suppose $\psi_1$ evolves to $\psi_2 = U(t)\psi_1$, as shown in Figure 1.

Figure 1: Commutative diagram: $|\psi_1\rangle$ evolves to $|\psi_2\rangle$

$$\begin{tikzcd}
|\psi_1\rangle \arrow[r, "U"] \arrow[d, "V"] & |\psi_2\rangle \arrow[d, "V"] \\
|\psi_1'\rangle \arrow[r, "U"] & |\psi_2'\rangle
\end{tikzcd}$$

$$\begin{aligned}
|\psi_1\rangle \xrightarrow{U} |\psi_2\rangle &\text{ or, equivalently} \\
\psi_2 = U\psi_1&\text{. Now, if } V \text{ really is a symmetry:} \\
|\psi_1'\rangle \xrightarrow{U} |\psi_2'\rangle &\text{ , or} \\
\psi_2' = U\psi_1'&\text{, so from (1) and Definition 1:} \\
V|\psi_2\rangle = UV|\psi_1\rangle &\text{ , or, using (3)} \\
VU|\psi_1\rangle = UV|\psi_1\rangle &\text{ . But } |\psi_1\rangle \text{ is an arbitrary state, so} \\
VU = UV&\text{. Hence for small time } \epsilon \\
V(I - i\epsilon H) = (I - i\epsilon H)V&\text{, whence} \\
VH = HV&\text{, which is (2)}
\end{aligned} \tag{3}$$

Moreover the symmetry $V$ should transform mutually exclusive states into mutually exclusive states, whence it should preserve orthogonality; $V$ should be unitary. □

Symmetries can be discrete or continuous.

- Discrete
  - reflection
  - interchange particles
- Continuous
  - rotation
  - translation

All continuous symmetries can be generated by $I - i\epsilon G$, for some Hermitean $G$ such that $[H, G] = 0$.

E.g.

$$\begin{aligned} V\psi(x) =& \psi(x-\epsilon), \text{ shift right} \\ =& \psi(x) - \epsilon \frac{\partial \psi}{\partial x} \\ V =& I - \epsilon \frac{\partial}{\partial x} \\ =& I - \frac{i\epsilon}{\hbar} P \\ G =& \frac{P_x}{\hbar}, \text{ Generator of } x \text{ translation} \end{aligned}$$

# 2 Symmetry groups and degeneracy

## 2.1 Symmetry groups and degeneracy

Symmetries are operations that you can do on a system which don't change:

- the description;
- the phenomena;
- the energy levels, and value of the energy.

Examples of symmetry:

- translation–doesn't change energy levels, Hamiltonian, or the Schroëdinger equation;
- rotation;
- interchanging identical particles;
- crystal symmetry: translate one lattice spacing (not exact if crystal not infinite).

Some symmetries are more abstract than others; the more obvious symmetries come from the properties of space; homogeneity and isotropy.

**Definition 3** (Degeneracy of energy levels)**.** *If there is more than one state with a given energy level, that energy level is called degenerate. We will see that it only happens when there is a symmetry: symmetries sometimes imply degeneracy, but not always.*

Degeneracy isn't a coincidence: symmetry sometimes implies degeneracy.

One application of symmetry is to analyze the energy of a system and see if it has energy levels that exactly match.

We will start with rotation symmetry and think about a very simple system, a particle moving in a circle. If it can't leave the circle we can describe its position using an angle $\theta$. It has a wave function $\psi(\theta)$, such that $\psi^*(\theta)\psi(\theta)$ gives the probability of finding the particle at a particular point.

For a small counter-clockwise rotation:

$$\begin{aligned}
\psi(\theta) \to &\psi(\theta - \epsilon) \\
\delta\psi = &-\epsilon\frac{\partial\psi}{\partial\theta}, \text{ c.f. linear momentum} \\
= &-i\epsilon\big(-i\frac{\partial\psi}{\partial\theta}\big), \text{ now defining the Angular Momentum operator } L\text{:} \\
L \triangleq &-i\hbar\frac{\partial}{\partial\theta}
\end{aligned}$$

We see that $L$ is Hermitean–c.f. $P = -i\hbar\frac{\partial}{\partial x}$

$$\delta\psi = -\frac{i\epsilon}{\hbar}L\psi, \text{ is the generator of rotation} \tag{4}$$

What are the Eigenvalues and Eigenvectors of Rotation?

$$\begin{aligned}
L\left|\psi\right\rangle =& m\left|\psi\right\rangle \\
-i\hbar\frac{\partial\psi}{\partial\theta} =& m\psi \\
\psi(\theta) =& e^{\frac{im\theta}{\hbar}}
\end{aligned}$$

Now, we want $\psi$ single valued (symmetry!), i.e.

$$\begin{aligned}
\psi(\theta + 2\pi) =& \psi(\theta), \text{ so} \\
\frac{m}{\hbar} =& k \in \mathbb{Z}. \text{ By convention, redefine } m \\
L =& m\hbar
\end{aligned} \tag{5}$$

This is the quantization of angular momentum.

**Definition 4** (Magnetic quantum number)**.** *The eigenvalue of $L$, $m$ in (5) is known as the Magnetic quantum number.*

We expect that the energy will depend on the angular momentum. Now $m \neq 0 \implies E(m) = E(-m)$, so we have degeneracy. A magnetic field breaks this; rotational symmetry isn't enough for degeneracy, but adding reflection symmetry is sufficient (but need two non-commuting symmetries).

We will use $M$ to denote reflection symmetry. If we have reflection symmetry, $E(m) = E(-m)$, because the reflection of a system with specified angular momentum is a system with opposite angular momentum (Avoid calling reflection symmetry "mirror symmetry", as this is used in string theory with a precise meaning).

The reflection of a magnetic field is not the same field, as it is an axial vector. Imagine a current generating a magnetic field going into the plane. If we reflect, the current goes in the opposite direction, as does the magnetic field.

**Theorem 5** (Commutator of two symmetries is itself a symmetry)**.**

$$\begin{aligned} \textit{if } [A, H] =& 0, \textit{and} \\ [B, H] =& 0, \textit{ and} \\ [A, B] =& iC, \textit{ then} \\ [C, H] =& 0 \end{aligned}$$

*Proof.*
**Case 1:** $C$ is a linear combination of $A$ and $B$; the result is trivial.
**Case 2:** $A$, $B$, and $C$ are independent

$$\begin{aligned} i[C, H] =& [A, B]H - H[A, B], \text{ expanding commutator} \\ =& ABH - BAH - HAB + HBA, \text{ again} \\ =& ABH - BAH - AHB + BHA, \text{ using } [A, H] = 0 \wedge [B, H] = 0 \\ =& \cancel{ABH} - \cancel{BAH} - \cancel{ABH} + \cancel{BAH}, \text{ using commutativity again and canceling} \\ =& 0 \end{aligned}$$

□

**Theorem 6** (Two symmetries that don't commute imply degeneracy)**.** *Suppose we have two symmetries, whose generators are $A$ and $B$.*

$$\begin{aligned} [A, H] =& 0, \textit{ and} \\ [B, H] =& 0, \textit{ and} \\ [A, B] \neq& 0 \end{aligned}$$

*then there are two or more independent eigenvectors of $H$ with the same eigenvalue (energy).*

*Proof.* Choose any eigenvector $|v_1\rangle$ of $H$ with eigenvalue $E_1$. Then we can write

$$\begin{aligned} H|v_1\rangle =& E_1|v_1\rangle \\ AH|v_1\rangle =& E_1 A|v_1\rangle \text{, since } E_1 \text{ is a scalar} \\ HA|v_1\rangle =& EA|v_1\rangle \text{, since } A \text{ commutes with } H \end{aligned}$$

Whence $A|v_1\rangle$ is also an eigenvector of $H$ with eigenvalue $E_1$. Similarly $B|v_1\rangle$ is also an eigenvector with eigenvalue $E_1$.
**Case 1:** If $A|v\rangle$, or $B|v\rangle$ is independent of $|v\rangle$, and of each other, we have constructed a pair of eigenvectors that share the same eigenvalue, and the result is proved.
**Case 2:** Otherwise choose another eigenvector $|v_2\rangle$. Continue ($|v_i\rangle$) until we have $A|v_i\rangle$, and $B|v_i\rangle$ independent as in the first case.
**Case 3:** If we exhaust all eigenvectors, then uses $\{v_i\}$ as a basis to diagonalize $A$ and $B$ simultaneously, so they commute. But this is impossible, by hypothesis. □

**Theorem 7** (Reflection and rotation don't commute)**.** *If $M\psi(\theta) = \psi(-\theta)$, $ML \neq LM$*

*Proof.*

$$\begin{aligned} MLe^{im\theta} =& Mme^{im\theta} \\ =& me^{-im\theta} \\ LMe^{im\theta} =& Le^{-im\theta} \\ =& -me^{-im\theta} \\ [M,L]e^{im\theta} =& 2me^{-im\theta} \end{aligned}$$

□

**Definition 8** (Commutator algebra)**.** *Closure of $\{A, B, C, ...\}$ under commutation.*

A collection of generators is an algebra.

**Definition 9** (abelian)**.** *A group of symmetries that all commute.*

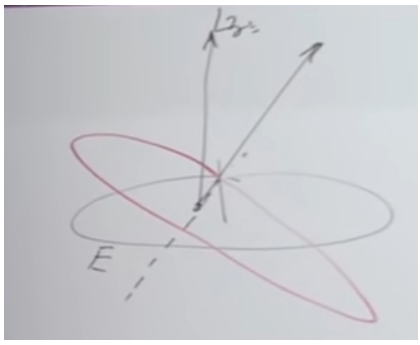
**Definition 10** (non-abelian)**.** *A group of symmetries that isn't abelian.*

## 2.2 An example: Angular Momentum

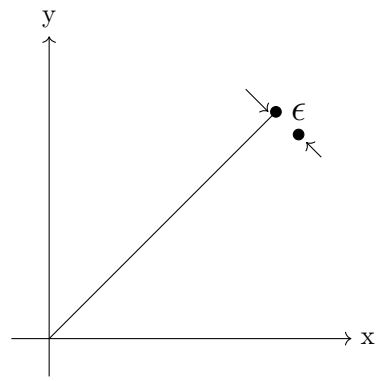
Consider a classical orbit with energy $E$, Figure 2a. Is it degenerate? All we have to do is rotate it to see a new orbit with energy $E$: rotation about the y-axis has added a z-component. This is a classical suggestion that the quantum mechanical operators of angular momentum don't commute (rotation about $2^{nd}$ axis converts rotation about $1^{st}$ into something with a component along $3^{rd}$).

Figure 2: An example: Angular Momentum

(a) Degeneracy of Angular Momentum(classical)




(b) Rotation in 2D by a small angle $\epsilon$




Let's look at the angular momentum generators. Figure 2b shows a rotation about the z-axis by a small angle $\epsilon$.

$$
\begin{aligned}
\delta x =& -y\epsilon \\
\delta y =& x\epsilon. \text{ Now for a wave function } \psi(x,y) \\
\delta\psi =& \frac{\partial\psi}{\partial x}\delta x + \frac{\partial\psi}{\partial y}\delta y \\
=& \big(-y\frac{\partial\psi}{\partial x} + x\frac{\partial\psi}{\partial y}\big)\epsilon \\
=& i\,\epsilon\big(-yP_x + xP_y x\big). \text{ If we define} \\
L_z =& xP_y - yP_x, \text{ we can write:} \\
\delta\psi =& i\epsilon L_z\psi. \text{ Similarly we define} \\
L_x =& yP_z - zP_y, \text{ and} \\
L_y =& xP_x - xP_z, \text{ c.f. the classical formula} \\
\vec{L} =& \vec{r}\times\vec{P}
\end{aligned}
$$

These are symmetries if the system is rotationally invariant.

Do they commute with each other? In [2] we saw

$$[L_i, H] = 0, \text{ for } i \in \{x, y, z\} \tag{6}$$
$$[L_x, L_y] = iL_z \tag{7}$$
$$[L_y, L_z] = iL_x \tag{8}$$
$$[L_z, L_x] = iL_y \tag{9}$$

**Definition 11** (Algebra)**.** *An* algebra over a field *(often simply called an algebra) is a vector space equipped with a bilinear product.*

**Definition 12** (Lie algebra)**.** *A Lie algebra is an algebra where multiplication is commutation. LS uses the definition that it is a collection of generators that is closed under commutation.*

So $L_x$, $L_y$, $L_z$ generate a Lie algebra: if we continue commuting we don't find anything new.

We want to find eigenvalues of angular momentum. It us useful to introduce creation and annihilation operators for $L_z$.

$$L_\pm \triangleq L_x \pm iL_y \tag{10}$$

$$[L_\pm, L_z] = \mp L_\pm \tag{11}$$

**Theorem 13** (If $|m\rangle$ is an eigenvector of $L_z$, so are $L_+ |m\rangle$ and $L_- |m\rangle$)**.** *If*

$$\begin{aligned} L_z |m\rangle =& m |m\rangle \text{, then} \\ L_z(L^+ |m\rangle) =& (m+1)(L^+ |m\rangle)\text{, and} \\ L_z(L^- |m\rangle) =& (m-1)(L^- |m\rangle) \end{aligned}$$

*Proof.* Suppose we have found one eigenvector of $L_z$:

$$L_z |m\rangle = m |m\rangle \text{, magnetic quantum number} \tag{12}$$

$$\begin{aligned} [L_+, L_z] |m\rangle =& \big(L_+L_z - L_zL_+\big) |m\rangle \\ =& -L_+ |m\rangle \text{, from (11). Rearranging and using (12)} \\ \underbrace{L_+L_z |m\rangle}_{mL_+|m\rangle} - L_zL_+ |m\rangle =& -L_+ |m\rangle \\ mL_+ |m\rangle + L_+ |m\rangle =& L_zL_+ |m\rangle \text{, whence} \\ \big(m+1\big)L_+ |m\rangle =& L_zL_+ |m\rangle \end{aligned} \tag{13}$$

So $L_+ |m\rangle$ is an eigenvector, eigenvalue $m+1$. Similarly, $L_- |m\rangle$ is an eigenvector, eigenvalue $m-1$. The sequence of eigenvectors terminates if $L_\pm |m\rangle = 0$. □

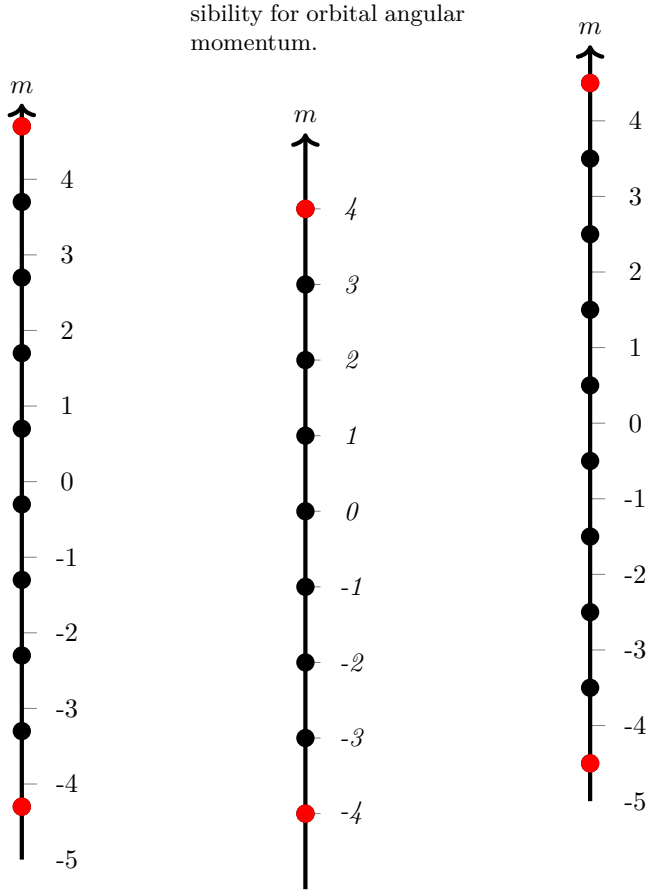
- So, using the algebra of commutators, we have generated a spectrum of values of $L_z$, separated by integers–see Figure 3a. This spectrum does not necessarily have to go on indefinitely, as it is possible that $\exists |m\rangle : L_+ |m\rangle = 0$.
- Imagine that we rotate by 180°: this takes $L_z \to -L_z$, so rotational symmetry requires that the termination points be reversed: they must be symmetrical. There are two possibilities–Figures 3b and 3c. We can show that the only values that are allowed for orbital angular momentum are integral (single values wave function), but half integral is also possible for spin.

Figure 3: Spectrum of $L_z$. Eigenvalues are separated by integers, and the red circles show possible terminations, where $L_\pm |m\rangle = 0$. Rotational symmetry implies that maximum/minimum values are symmetrical about zero.

(a) We have not yet shown that the eigenvalues are necessarily integers.

(b) Integral values, exhibiting symmetry about 0. This is the only possibility for orbital angular momentum.

(c) Half integral values: this is possible for spin.




We now show that the multiplet of states in Figure 3b all have the same energy.

**Theorem 14** (Eigenvectors of $L_z$ are degenerate eigenvectors of $H$)**.**

$$L_z |m\rangle = m |m\rangle \wedge H |m\rangle = E |m\rangle$$
$$\Longrightarrow$$
$$H |m \pm 1\rangle = E |m \pm 1\rangle$$

*Proof.*

$$\begin{aligned} H\left|m\right\rangle =& E\left|m\right\rangle \\ HL_+\left|m\right\rangle =& L_+H\left|m\right\rangle \text{, from (6), } H \text{ commutes with all } L \\ =& L_+E\left|m\right\rangle \text{, from (14)} \\ =& EL_+\left|m\right\rangle \\ H\left|m+1\right\rangle =& E\left|m+1\right\rangle \text{, from (13)} \end{aligned} \tag{14}$$

So we have a different state which is an eigenvector with the same eigenvalue. □

# 3 Atomic orbits and harmonic oscillators

## 3.1 Atomic orbits and Angular Momentum

We will talk more about what angular momentum is about.

If a particle moves in a central force field, angular momentum and, hence, the orbital plane are preserved. The state is $\psi(r,\theta,\phi) = \psi(r,\theta)$; if system were 2 dimensional, the angular momentum would be:

$$\begin{aligned} L =& -i\frac{\partial}{\partial\theta}\text{, and the eigenvalues and eigenvectors would satisfy} \\ -i\frac{\partial\psi(r,\theta)}{\partial\theta} =& l\psi(r,\theta)\text{, which has solution} \\ \psi(r,\theta) =& e^{il\theta}\chi(r)\text{, for some } \chi \end{aligned}$$

In 3 dimensions, $\psi(r,\theta,\phi) = Y(\theta,\phi)\chi(r)$, where $Y(\theta,\phi)$ encapsulates the angular dependence of the wave function.

In Section 2 we derived the commutators, (7), (8), and (9). We selected one component, $L_z$, and worked with its eigenvectors, $L_z\left|m\right\rangle = m\left|m\right\rangle$. We defined $L_\pm$, (10), and found that they had useful properties, (11). $L_\pm$ were raising and lowering operators, which take use up and down the spectrum.

We asked whether the raising and lowering could go on forever. The only way to come to an end is if $\exists l$ such that $L_+\left|k\right\rangle = 0$ (and similarly for $L_-$). The spectrum has to be symmetric (rotational invariance). We found that the spectrum had to go up in integers, and that it started at either an integer or half integer. We found that the spectrum was a multiplet of $2l+1$ states. The multiplet is characterized by the magnitude of angular momentum.

From (13) we have a spectrum of angular momenta, integral or half integral, say $\{\,-l,...,l : L_+ \mid l\rangle = 0\,\}$. There are $2l+1$ states with constant $L^2 = L_x^2 + L_y^2 + L_x^2$. Classically $L^2 = L_z^2 + (L_x - iL_y)(L_x + iL_y)$, but this fails in quantum mechanics as they operators don't commute.

The quantum version is:

$$\begin{aligned}
L^2 =& L_z^2 + (L_x - iL_y)(L_x + iL_y) - i[L_x, L_y] \\
=& L_z^2 + L_z + L^- L^+ \\
L^2 \left|l\right\rangle =& L_z^2 \left|l\right\rangle + L_z \left|l\right\rangle + L^- L^+ \left|l\right\rangle \\
=& l^2 \left|l\right\rangle + l \left|l\right\rangle + 0, \text{ because } \left|l\right\rangle \text{ is an eigenvector \& } L^+ \left|l\right\rangle = 0. \\
L^2 \left|l\right\rangle =& l(l+1) \left|l\right\rangle \qquad (15)
\end{aligned}$$

**Theorem 15** (Degeneracy of $L^2$)**.** *1.* $[L^2, L_i] = 0$

*2. All eigenvectors of $L_z$ are eigenvectors of $L^2$, eigenvalue $l(l+1)$*

*Proof.* Part 1

$$\begin{aligned}
[L^2, L_x] =& [L_x^2 + L_y^2 + L_2^2, L_x] \\
=& \cancel{[L_x^2, L_x]} + [L_y^2, L_x] + [L_z^2, L_x] \qquad (16) \\
[L_y^2, L_x] =& L_y L_y L_x - L_y L_x L_y + L_y L_x L_y - L_x L_y L_y \\
=& - L_y [L_x, L_y] - [L_x, L_y] L_y \\
=& - iL_y L_z - iL_z L_y, \text{ using (7)} \qquad (17) \\
[L_z^2, L_x] =& L_z L_z L_x - L_z L_x L_z + L_z L_x L_z - L_x L_z L_z \\
=& L_z [L_z, L_x] + [L_z, L_x] L_z \\
=& iL_z L_y + iL_y L_z, \text{ using (9)} \qquad (18) \\
[L^2, L_x] =& 0, \text{ on substituing (17) and (18) in (16)}
\end{aligned}$$

Part 2. We know from (15) that the eigenvector of $L_z$ with the maximum eigenvalue satisfies the theorem. The theorem follows by induction once we have the following Lemma.

**Lemma 16** (Induction step for Theorem 15)**.** *If $\left|m\right\rangle$ is an eigenvector of $L_z$, and it is also an eigenvector of $L^2$, with eigenvalue $l(l+1)$, and $L_- \left|m\right\rangle$ is not zero, then $L_- \left|m\right\rangle$ is an eigenvector of $L^2$, with eigenvalue $l(l+1)$.*

*Proof.*

$$\begin{aligned}
L^2 \left|m\right\rangle =& l(l+1) \left|m\right\rangle, \text{ by hypothesis. Now} \\
[L_-, L^2] =& 0, \text{ from (10) and Part 1, hence} \\
L^2(L_- \left|m\right\rangle) =& (L^2 L_-) \left|m\right\rangle \\
=& (L_- L^2) \left|m\right\rangle \\
=& L_-(L^2 \left|m\right\rangle) \\
=& L_-[l(l+1) \left|m\right\rangle] \\
=& l(l+1)(L_- \left|m\right\rangle)
\end{aligned}$$

□

□

When we find a multiplet such as Figure 3a, we not only have eigenvectors of $L_z$, but also of $L^2$. For each $l$, there are $2l+1$ states with the same $L^2$, and the same Energy[2]. Classically they are just tilts–Figure 2a.

We have a collection of functions of the angles, characterized by $m$ and $l$, $Y_{ml}(\theta, \phi)$. They are known as "spherical harmonics", and they are the analogues on the unit sphere of $e^{\pm il\theta}$

## 3.2 The Central Force Problem

We will use classical mechanics to guess a solution.

Classically:

$$H = \frac{\vec{P}^2}{2m} + V(r), \text{ conserved} \tag{19}$$

$$\vec{L} = \vec{r} \times \vec{P}, \text{ angular momentum is conserved, so use } xy \text{ plane}$$

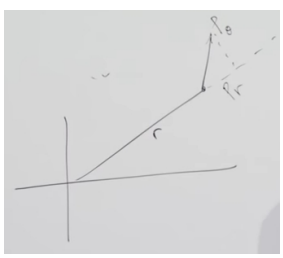
$$H = \frac{P_r^2 + P_\theta^2}{2m} + V(r), \text{ resolving as shown in Figure 4a} \tag{20}$$

$$= \frac{P_r^2}{2m} + \frac{L^2}{2mr^2} + V(r), \text{ since } |L| = |r|\,|P|$$

So we have an equation for a one dimensional problem. Figure 4b depicts the potential, and Figure 4c shows a solution, oscillations about the $r$ that yields the minimum potential.

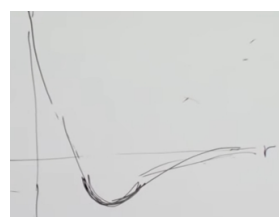
Figure 4: Central Force as a one dimensional problem

(a) Resolving momentum into radial and angular components



(b) Potential assuming the Coulomb force

(c) Small oscillations about minimum



That is the classical physics. For the quantum physics we have a wave function which satisfies the Schrödinger Equation(21). The angular part is

[2]Because $[L_i, H] = 0$–(6)

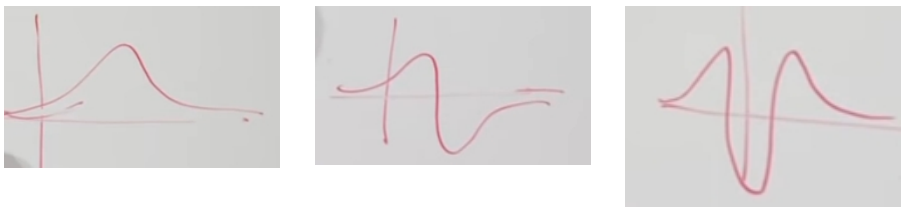
totally taken care of by our existing study of angular momentum[3].

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(r)}{\partial r^2}+\hbar^2\frac{l(l+1)\hbar^2}{r^2}\psi(r)+V(r)\psi(r)=E\psi(r) \tag{21}$$

Figure 4b depicts the potential for the Coulomb force. The energy levels of (21) are characterized by the number of nodes–Definition 17 and Figures 5a, 5b, and 5c (The more nodes, the faster the wiggle, hence higher momentum).

Figure 5: Solving Central Potential Schrödinger Equation (21)

(a) No nodes (b) One node (c) Two nodes



**Definition 17** (Node)**.** *A node is a point where the wave function is zero.*

**Theorem 18** (Nodes and energy levels)**.** *The ground state has 0 nodes, the first excited one node, etc.*

What can we say about energy levels in general? Figure 6 plots the number of energy levels against $l$, and plots number of nodes vertically. It exhibits degeneracy. If the energy is too high the electron escapes. The Coulomb potential has a very special feature. It is almost an accident: it is violated by the finite size of the nucleus, by relativistic corrections, by spin. *The zero node energy for $l=1$ is equal to the 1 node energy for $l=0$, etc*–Figure 7.

[3]Strictly step (20) requires a quantum mechanical justification

Figure 6: Degeneracy of Energy Levels in Hydrogen Atom. Each $l$ has its own Schroëdinger equation (21), so each $l$ may have its own set of bound solutions. The ground state for $l = 0$ has no nodes, the next state has one node, etc. For $l = 1$ the energy of the ground state lies somewhere above the corresponding state for $l = 1$. Moreover there are 3 states for each node: the energy levels are degenerate. Similarly for $l = 2$ there are 5 states for each node, and the energy levels are higher. NB: if $E$ high enough, electron escapes.

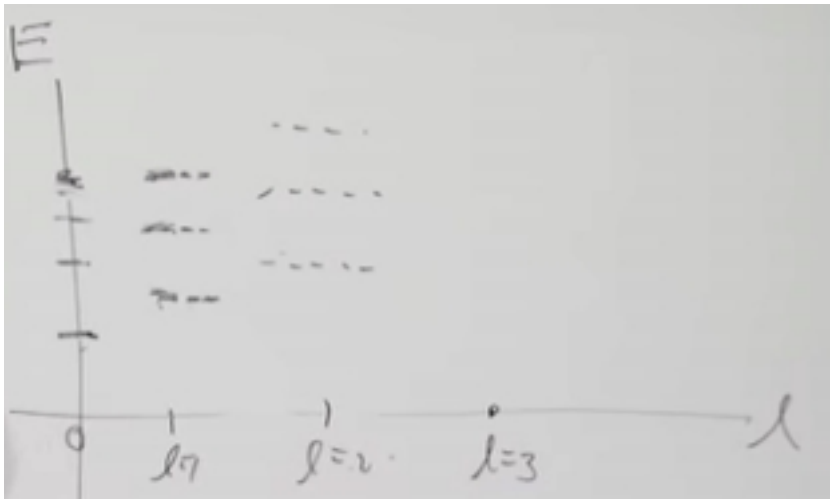
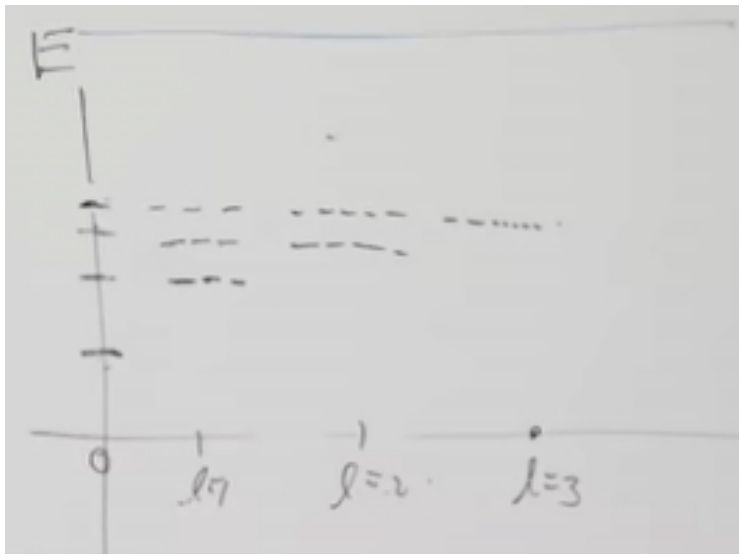


Figure 7: For the Coulomb force *only*, the energy levels for $l+1$ match those for $l$, shifted up by one. This is an extra degeneracy from a bizarre symmetry. The number of states at each level is $\{1, 4, 9, 16, ...\}$. NB, the symmetry is violated for every real atom, because of correction terms (spin, etc).




Real atoms behave a bit differently.

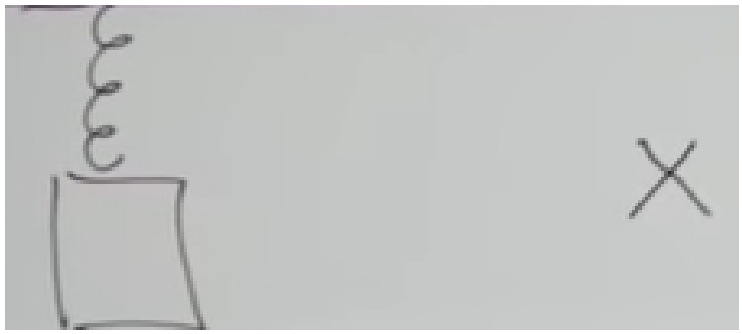
- the nucleus has finite size, so at short distances the Coulomb law isn't quite accurate;
- Instead of $\{1, 4, 9, 16, ...\}$ states at each energy level there are $\{2, 8, 18, 32, ...\}$–we have ignored *spin*.

Are there always raising an lowering operators? No: only when you are very lucky; physics has been lucky twice. The harmonic oscillator piece of luck is very pervasive, as many things in nature can be well approximated by a harmonic oscillator. For example, go to a higher momentum state of the central force problem, Figure 4c, and the simplest solution is to treat them as a harmonic oscillator.

## 3.3 Harmonic oscillators

Everything in physics that has an equilibrium, if you disturb equilibrium by a small amount, the behaviour can be approximated by a harmonic oscillator. The model is just a suspended mass ($m = 1$), spring constant ($k = \omega^2$)

Figure 8: Suspended mass ($m = 1$), spring constant ($k = \omega^2$). $X$ is deviation from equillibrium.




Classically:

$$\begin{aligned}
H =& \frac{p^2}{2} + \frac{\omega^2 x^2}{2}. \text{ We'll check with Hamilton's equations} \\
\dot{x} =& \frac{\partial H}{\partial p} \\
=& p \\
\dot{p} =& -\frac{\partial H}{\partial x} \\
=& -\omega^2 x \\
\ddot{x} + \omega^2 x =& 0, \text{ as expected.}
\end{aligned}$$

Quantum mechanically:

$$\begin{aligned}
H =& \frac{P^2}{2m} + \frac{\omega^2 X^2}{2} \\
=& \frac{1}{2}\big(P + i\omega X\big)\big(P - i\omega X\big) - \frac{i\omega}{2}[X, P], \text{ since } X \text{ and } P \text{ don't commute} \\
=& \frac{1}{2}\big(P + i\omega X\big)\big(P - i\omega X\big) + \underbrace{\frac{\hbar\omega}{2}}_{\text{zero point energy}}
\end{aligned}$$

Classically both $p = 0$ and $x = 0$ in the ground state; since the Heisenberg uncertainty principle precludes $p = 0$ and $x = 0$, the zero point energy is the minimum possible. We'll drop the ground state energy (zero point energy) for the time being, since it is just an additive constant[4].

$$H = \omega \frac{\big(P + i\omega X\big)}{\sqrt{2\omega}} \frac{\big(P - i\omega X\big)}{\sqrt{2\omega}} \tag{22}$$

[4] At the end of lecture 10, LS mentioned that the zero point energy is involved only in gravitational interactions; elsewhere only the energy *difference* matters

We'll introduce raising and lowering operators, to perform the same task as they did for angular momentum. They will take us up and down a ladder, this time a ladder of energy levels.

$$a^+ \triangleq \frac{P + i\omega x}{\sqrt{2\omega}} \tag{23}$$

$$a^- \triangleq \frac{P - i\omega x}{\sqrt{2\omega}} \tag{24}$$

$a^- = (a^+)^\dagger$, the operators are Hermitean conjugates and (22) becomes:

$$H = \omega a^+ a^- \tag{25}$$

We'll take the commutator:

$$\begin{aligned} [a^-, a^+] =& \frac{1}{2\omega}\left[P - i\omega x, P + i\omega x\right] \\ =& 1. \text{ We also define} \qquad (26) \\ N \triangleq& a^+ a^-, \text{ so (25) becomes} \qquad (27) \\ H =& \omega N \end{aligned}$$

$N$ is Hermitian, so it has a complete set of eigenvalues and eigenvectors. Let $|n\rangle$ and $n$ be any pair of an eigenvector and eigenvalue (we aren't *assuming* that $n$ is an integer).

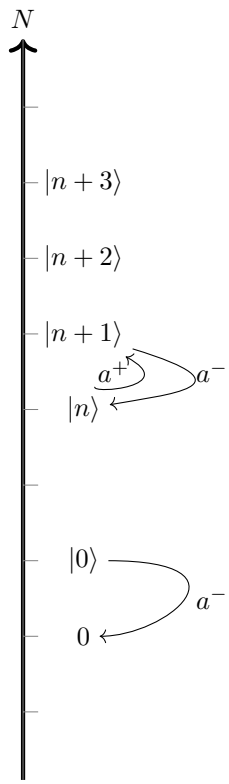
$$\begin{aligned} N|n\rangle =& n|n\rangle \qquad (28) \\ a^+ \underbrace{(a^- a^+ - a^+ a^-)}_{=1} |n\rangle =& a^+|n\rangle, \text{ using (26)} \\ a^+ a^- a^+ |n\rangle - a^+ \underbrace{a^+ a^- |n\rangle}_{n|n\rangle} =& a^+|n\rangle \\ \underbrace{a^+ a^-}_{N} a^+ |n\rangle =& n a^+ |n\rangle + a^+ |n\rangle, \text{ applying (28) and (27).} \\ N a^+ |n\rangle =& (n+1) a^+ |n\rangle \end{aligned}$$

So $a^+$ acts as raising operator; we now show $a^-$ is lowering.

$$\begin{aligned} a^-\left(a^+ a^- - a^- a^+\right)|n\rangle =& -a^-|n\rangle \\ \underbrace{a^- a^+}_{\mathrm{N}} a^- |n\rangle - a^- \underbrace{a^- a^+}_{\mathrm{N}} |n\rangle =& -a^-|n\rangle \\ N a^- |n\rangle =& (n-1) a^- |n\rangle \end{aligned}$$

Since the Hamiltonian is positive we can't ever get negative energy, so there must be a lowest energy state, $|0\rangle$–Figure 9. In QM there is a theorem that all the eigenvalues of $MM^\dagger$ are positive or zero, for any operator $M$.

Figure 9: Energy Spectrum for Harmonic Oscillator. $a^+$ takes energy up one level, and $a^-$ down, except for $a^- |0\rangle$=0.




We will see (Theorem 19) the appropriate normalization gives:

$$a^+ |n\rangle = \sqrt{n+1}\, |n+1\rangle$$
$$a^- |n\rangle = \sqrt{n}\, |n-1\rangle$$

# 4 Spin

## 4.1 Harmonic oscillators(continued)

Harmonic oscillators are ubiquitous. Any time a system has an equilibrium, there can be oscillations about equilibrium. An oscillator can oscillate at different frequencies simultaneously (harmonics)–e.g. a violin string. We will be particularly interested in oscillations in an electromagnetic field. For example, if you have radiation in a cavity, there will be a variety of oscillations. What is the equilibrium? No electromagnetic field, i.e. vacuum is the equilibrium

for electromagnetic field. Give it a knock, e.g. microwaves, and there will be oscillations.

In a crystal, the oscillations of one atom affect its neighbours, so we have a field. The quanta are called phonons.

In Section 3.3 we found:

$$\begin{aligned}
H =& \frac{P^2}{2m} + \frac{\omega^2 x^2}{2}, \text{ and we defined} \\
a^{\pm} =& \frac{q \pm i\omega x}{\sqrt{2\omega}}, \text{ raising and lowering operators} \\
[a^-, a^+] =& 1 \\
N =& a^+ a^-, \text{ whence} \\
H =& \omega(N + \frac{1}{2})
\end{aligned}$$

We will find ground state by solving the Schrödinger equation.

$$\begin{aligned}
|\psi\rangle \to& \psi_0(x), \text{ in the ground state} \\
X \to& x, \text{ multiplication} \\
P \to& -i\frac{d}{dx}
\end{aligned}$$

We know that there has to be a ground state. We can't use $a^-$ indefinitely; since $H$ is positive, there can be no negative eigenvalues. So $\exists\, |0\rangle$ such that:

$$\begin{aligned}
a^- |0\rangle =& 0 \\
N |0\rangle =& a^+ a^- |0\rangle \\
=& a^+ 0 \\
=& 0
\end{aligned}$$

NB $|0\rangle \neq 0$. $|0\rangle$ is a vector which can be normalized; 0 is a vector of length 0. Applying the transformations given above, we get the time independent Schrödinger equation.

$$-\frac{1}{2}\frac{d^2\psi(x)}{dx^2} + \omega^2 x^2 \psi(x) = E\psi(x)$$

There are solutions for all $E$, but the solutions are not normalizable unless E is in the spectrum, Figure 9. So we impose the condition $\langle\psi|\psi\rangle = 1$, or:

$$\begin{aligned}
\int_{-\infty}^{\infty} dx\psi^*(x)\psi(x) =& 1. \text{ We know that} \\
a^- \left|0\right\rangle =& 0, \text{ whence} \\
\big(-i\frac{d}{dx} - i\omega x\big)\psi_0(x) =& 0. \text{ Write} \\
\psi(x) =& e^{f(x)}, \text{ then} \\
\psi'(x) =& f' e^{f(x)} \\
i\big[f' + \omega x\big]e^{f(x)} =& 0 \\
f(x) =& -\frac{\omega x^2}{2} + C \\
\psi_0(x) =& e^{-\frac{\omega x^2}{2}}, \text{ not normalized.}
\end{aligned}$$

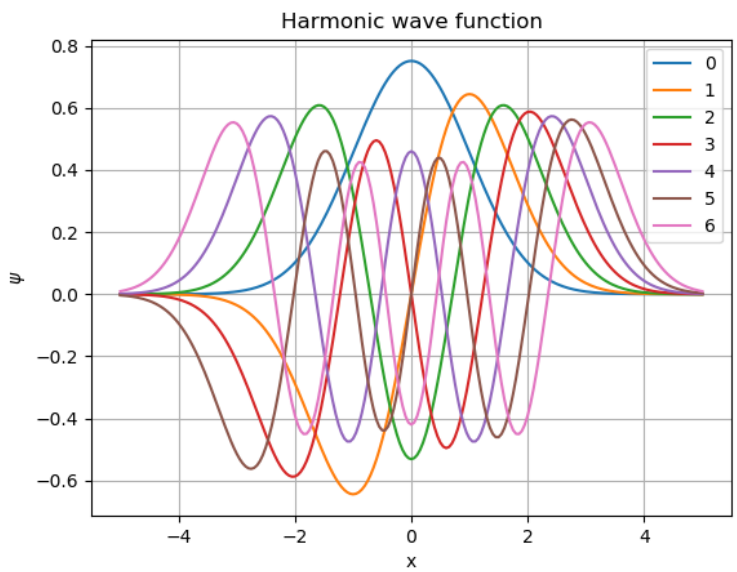
We can then use $a^+$ to generate excited states.

$$\begin{aligned}
\left|1\right\rangle =& a^+ \left|0\right\rangle, \text{ or, in the Schroëdinger representation:} \\
\psi_1(x) =& \big(-i\frac{d}{dx} + i\omega x\big)e^{-\frac{\omega x^2}{2}} \\
=& 2i\omega e^{-\frac{\omega x^2}{2}}
\end{aligned}$$

This is an odd function (antisymmetric)–Figure 10. It has one node.

Figure 10 shows how higher levels have more wiggles (momentum) and are away from origin (the particle is more likely to be far away) most of time.

Figure 10: Wave functions for harmonic oscillator




To see classical behaviour, superpose states to create wave packets. Time dependent looks like classical oscillator, especially at high energy (energy larger that ground state).

## 4.2 Spin

Some particles have half-integral spin, e.g. electron or proton. At rest they have no orbital rotation, no $\vec{r} \times \vec{p}$, only spin. Spin is a kind of angular momentum that is attached to a particle. You can think of it as an abstract quantity or as a tiny thing that is literally spinning. It transforms under rotation as angular momentum does. Spin has two states, up or down, left or right, in or out. The thing that characterizes angular momentum is having three components, which can be represented by matrices satisfying (7), (8), and (9). These are closely related to the commutation relations of the Pauli matrices (we will work in the representation of the $z$ component of the spin):

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$
$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

The Pauli matrices are Hermitian. We define:

$$\vec{s} = \frac{\vec{\sigma}}{2}$$

We can show the $s_i$ satisfy (7), (8), and (9).

$$\begin{aligned}[s_x, s_y] &= i s_z \\ [s_y, s_z] &= i s_x \\ [s_z, s_x] &= i s_y\end{aligned}$$

So the Pauli matrices, operating on 2D vectors, represent a simple angular momentum system, which is attached to the particles. Now we can ask about eigenvalues and eigenvectors. The Eigenvalues of $s_z$ are $\pm\frac{1}{2}$.

In Figures 3b and 3c we saw there were two possibilities for the spectrum of angular momentum, integral and half integral spins. We could have *guessed* the Pauli matrices by noticing the spin $-\frac{1}{2}, \frac{1}{2}$ case. Note: there are spin 0 particles, such as the Higgs, and the deuteron. We will largely focus on spin $\frac{1}{2}$ particles.

We define $J$, the total angular momentum:

$$J = L + s, \text{ where } L \text{ denotes the orbital angular momentum } \vec{r} \times \vec{P}$$

There are QM rules for adding angular momenta, which we won't cover in this lecture.

The first experimental background came from spectroscopy and the periodic table. The goal of this lecture is to see how periodic table emerges from spin. Pauli realized spin was needed.

$L^2 = l(l+1)$. We look at the energy levels of hydrogen –see Figure 6. There is a bunch of states for $l = 0$, for zero nodes($n = 0$), 1 node, etc. For $l = 1$ there are $2l + 1 = 3$ states for each node, and ground state for $l = 1$ has more energy than ground state for $l = 0$, because it has angular momentum.

Figure 7 shows the coincidence that the energy for for one value of $l$ matches that $l - 1$, except that energies are shifted in $n$: $E(n, l)$ satisfies $E(n, l + 1) = E(n + 1, l)$. So the number of states for each n is 1, 4, 9, 16, ...

The Helium nucleus has twice the charge of the hydrogen nucleus, so electrons are pulled in tighter, but the structure of Figure 7 is preserved for a Helium ion (negative). Put another electron in: we can put 2 electrons into ground state; if we try to create an ion with 3 electrons, one goes into first excited state!

## 4.3 Pauli exclusion principle

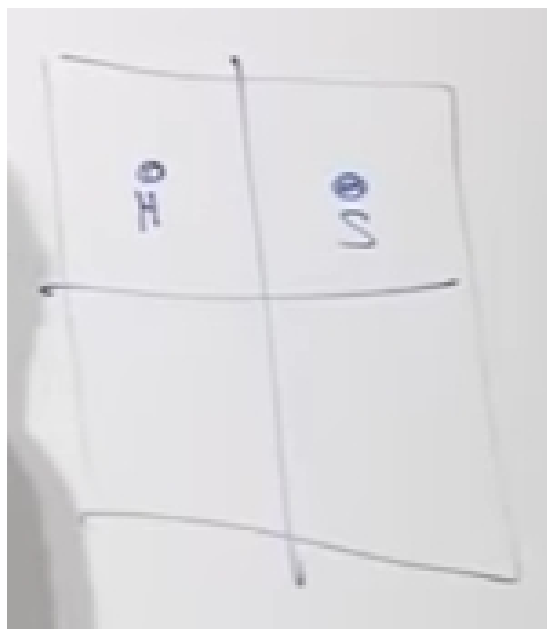
Pauli exclusion principle: no two electrons can ever get into the same state. But we can put 2 electrons into ground state. Pauli said there was a new property with two values, up and down.

Lithium has 3 electrons, 2 in ground state, 3rd in 1st excited state. There is room for 8 states in first excited level, if we allow for spin. This explains 2nd row of periodic table. The fact that spin was angular momentum was checked with magnetic field (split energy levels).

Pauli's Exclusion Principle is a *postulate* of non-relativistic QM, but a *consequence* of relativistic QM. There are two kinds of particles: ones that satisfy Pauli's Exclusion Principle (Fermions), and those that don't (Bosons).

In statistical mechanics, how do we handle identical particles? Imagine that we are putting particles into boxes (real or phase space). In Figure 11 we can imagine each particle having a name painted in "classical paint", so they are distinct. Or we can treat them as absolutely identical, so we usually count as if they were identical. In quantum mechanics, however, identical particles are indistinguishable.

Figure 11: Configurations Particles in Boxes: if we exchange labels, do we count as separate configurations?




Consider two particles, forgetting spin.

$$\begin{aligned}
\psi(x) &= \langle x|\psi\rangle \\
\psi(x_1, x_2) &= \langle x_1 x_2|\psi\rangle \text{ , two particles} \\
\psi(x_1, x_2) &\rightarrow \psi(x_2, x_1), \text{ swap using operator } S \\
S\,|x1, x2\rangle &= |x2, x1\rangle \\
S^2 &= 1
\end{aligned}$$

$S$ is unitary, so eigenvalues are $\pm 1$.

# 5 Fermions: a tale of two minus signs

We'll review the chemistry that went into the discovery of spin, and the Fermionic character of electrons. The Fermionic character implies two things that are related:

- spin of electron is half spin $\pm\frac{1}{2}$
- the electron satisfies the Pauli Exclusion Principle.

We can put 2 electrons into the ground state only because of spin. NB, the 2 electrons in Helium are *entangled*.

Photons don't have an exclusion principle; in fact they have a tendency to congregate.

Can we have spin $\frac{1}{2}$ particles that don't obey Pauli exclusion principle, or a spin integer that does obey Pauli exclusion principle? No. Can show this from quantum field theory.

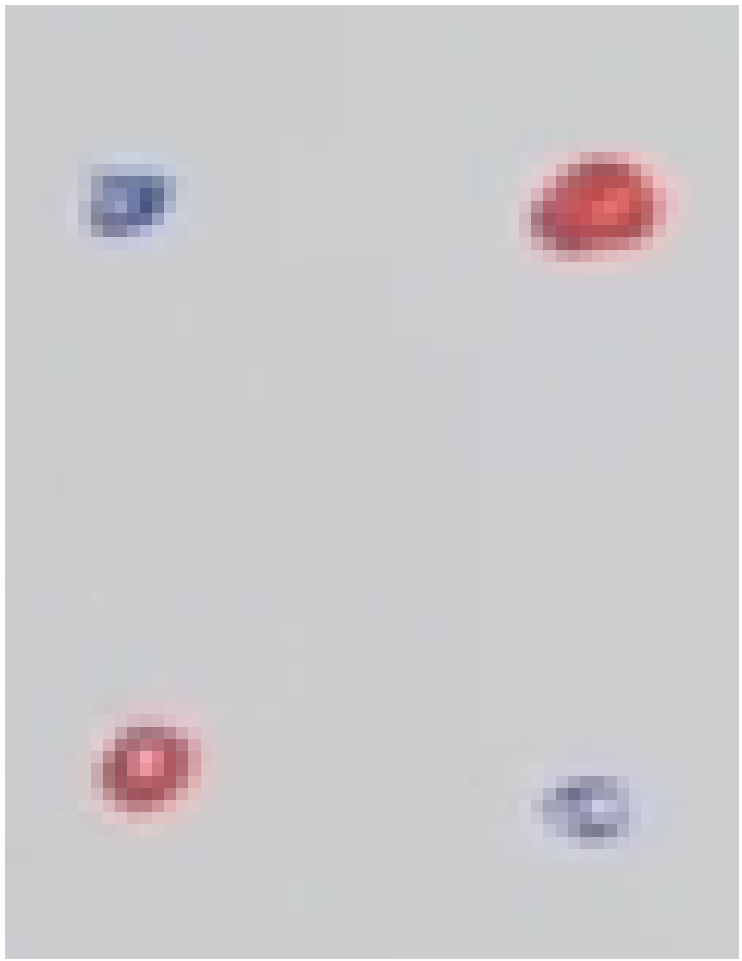
This lecture is a tale of two minus signs, one from exchange–Section 5.1, the other from spin–Section 5.2.

## 5.1 Swapping particles

The Pauli Exclusion Principle is deeper than simply saying that you cannot put two particles into the same state. It is actually a statement about multi particle wave functions. Let's forget about spin for the moment.

Consider wave function of positions for multiple particles, e.g. $|x_1, x_2, x_3\rangle$. Then $\langle x_1, ...x_n|\psi\rangle = \psi(x_1, ...x_n)$, and $\psi^*(x_1, ...x_n)\psi(x_1, ...x_n)$ is the probability that the system is in state $|x_1, ...x_n\rangle$. We ask whether $\psi(x_1, x_2, ...x_n)$ is the same as $\psi(x_2, x_1, ...x_n)$, or, equivalently, whether the upper configuration in Figure 12 is the same as the lower one. All experiments and theory agree that they are the same state: $|x1, x2\rangle = |x2, x1\rangle$ .

Figure 12: Is the upper configuration the same as the lower one?



On the other hand, the physical properties of a state do not depend on the overall phase. We can't say that $|x_1, x_2\rangle = |x_2, x_1\rangle$, as there may be a phase $e^{i\phi}$.

$$|x_1, x_2\rangle = |x_2, x_1\rangle\, e^{i\phi}, \text{ interchange twice–there are two possibilities}$$
$$|x_1, x_2\rangle = +\,|x_2, x_1\rangle \;\; \text{Bosons}$$
$$|x_1, x_2\rangle = -\,|x_2, x_1\rangle \;\; \text{Fermions}$$

NB: sign is not observable.

Table 1: Wave functions for Fermions and Bosons

| Wave function | Fermions | Bosons |
|---|---|---|
| $\psi(x_1, x_2) = -\psi(x_2, x_1)$ | OK | not OK |
| $\psi(x_1, x_2) = \psi(x_2, x_1)$ | not OK | OK |
| $\psi_0(x_1)\psi_0(x_2)$ | not OK | OK |
| $\psi_0(x_1)\psi_1(x_2)$ | not OK | not OK |
| $\psi_0(x_1)\psi_1(x_2) + \psi_0(x_2)\psi_1(x_1)$ | not OK | OK |
| $\psi_0(x_1)\psi_1(x_2) - \psi_0(x_2)\psi_1(x_1)$ | OK | not OK |

When there is spin we can think of the wave function as a function of position and spin–$\psi(x, \sigma_z)$ and $\psi(x1, \sigma_1, x2, \sigma_2, ...)$. You cannot put two Fermions into the same state.

## 5.2 Spin

Rotation by $2\pi$ is a mathematical process that you might expect to do nothing. Let's consider a wave function for one particle $\psi(x, \sigma)$ and rotate the coordinate system or the particle. What happens when you rotate a system with a given angular momentum by an angle about any axis–$R(\theta)2\pi$? What about $R(2\pi)2\pi$? Surely this leaves the system unchanged? But the system is analogous to swapping particles–Section 5.1. If all we want is for the probabilities and expectation to be preserved, perhaps the rotations introduce a phase–$\pm 1$.

Consider the generator of rotations:

$$J_z \left|\psi\right\rangle = -i\frac{\partial \left|\psi\right\rangle}{\partial\theta}$$

Imagine that angular momentum about $z$ has a definite value, so $\left|\psi\right\rangle$ is an eigenvector.

$$\begin{aligned} J_z \left|\psi\right\rangle =& m \left|\psi\right\rangle \text{, which is easily solved.} \\ \left|\psi(\theta)\right\rangle =& e^{im\theta} \left|\psi(0)\right\rangle \end{aligned}$$

But, from the commutation relations[5] we know that $m$ integral or half integral (in 3D, not 2D). What if $m$ half integral and we rotate by $2\pi$? We have phase of $-1$, so $\psi$ changes sign. Fermions follow $-1$, Bosons $+1$. Of course rotation by $4\pi$ does give the identity.

Leonard Susskind demonstrated that $2\pi \neq 0$:

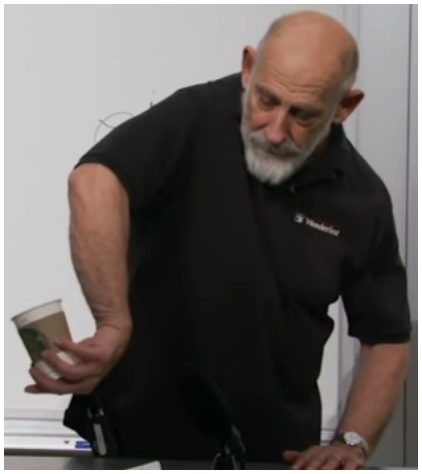
- Gerard 't Hooft Coffee Cup–Figure 13
- Dirac sphere in a box

[5] See discussion preceding Figure 3

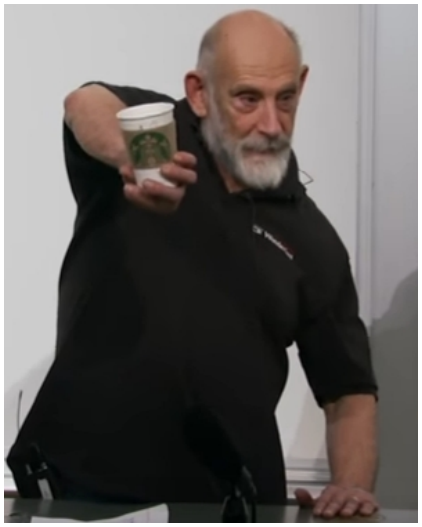
In each case we are following a path in rotation space.

Figure 13: The Gerard 't Hooft Coffee Cup trick

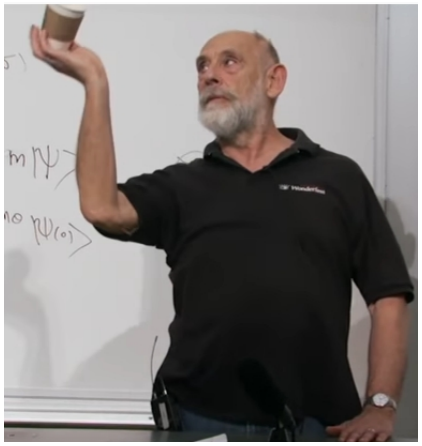
(a) Starting the first rotation



(b) $2\pi$: That is not the identity: let me tell you I'm quite certain that is not the identity!



(c) Staring the 2nd rotation



(d) $4\pi$: that is the identity

- The wave function for two fermions changes sign when we swap them;

- The wave function for a single fermion changes sign when we rotate by $2\pi$.

## 5.3 The connection between interchange and rotation

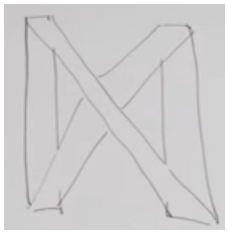
Is there a connection between interchange and rotation? Yes: it was discovered by David Finkelstein.

Figure 14: The David Finkelstein Belt Trick: A rotation by $2\pi$, plus an exchange, gives an untwisted belt

(a) Exchange of the two loops

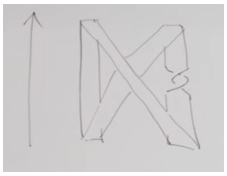
(b) Turn around and we see 2 rotations instead

(c) Not a possible configuration from twisting a belt

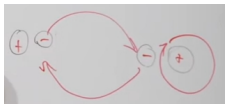


(d) The twist makes it legal: you can pull it apart to get the topologically trivial configuration.

(e) Add time, and create two pairs of particles at bottom



(f) Rotate one particle, and switch two of them. The switch and rotate give the same configuration



This shows the deep topological connection between rotation and interchange.

Imagine that I have two particles. I take one at rotate it by $2\pi$, then I switch two of them. Then, regardless of whether they are Fermions or Bosons, doing

nothing is equivalent to doing a rotation and an exchange–Figure 14f (this isn't the way Pauli did it).

This would not make sense if particles were eternal. Quantum Field Theory allows particles to be created. We start with no particles, then create particle-antiparticle pairs–Figure 14e. Do whatever it takes to create particles out of nothing. As time goes on, at the same time we rotate one particle and exchange the two negative ones. Topologically this is equivalent to just creating the two particles and not doing anything.

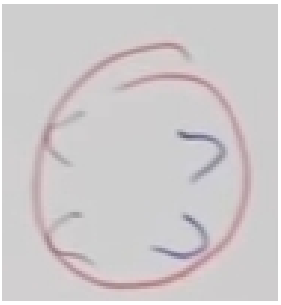
The spin statistics theorem is more general than the version Pauli proved. It is also true for solitons. A soliton is (solitary wave) is a lump of field (Finkelstein, Rubinstein). Its true for lumps of stuff in condensed matter physics, for manifolds, for all kinds of things where Pauli's argument doesn't apply.

Assume field is mostly zero, but there are lumps. The spin statistics theorem says that there are two kinds of lumps: those whose wave function changes sign when rotated, and those where it doesn't change–Figure 15–Fermions and Bosons.

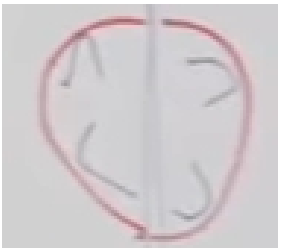
Figure 15: Annihilation of lumps and anti-lumps

(a) Blue and black lumps

(b) Mirror image



(c) Bring together - black lumps annihilated



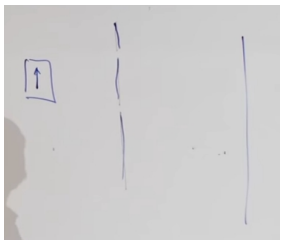
Is there a way to observe the change in phase of a Fermion?

Let's start with a experiment where you wouldn't see anything. Put electron in a cavity with a magnetic field, *slowly* rotate box, release, and generate interference pattern–Figure 16a. Compare with another experiment where we don't rotate: the interference pattern won't show any differences–Figure 16b. Now use a beam splitter, and we see interference from the $2\pi$ rotation (we don't know which box contains electron: don't look!).

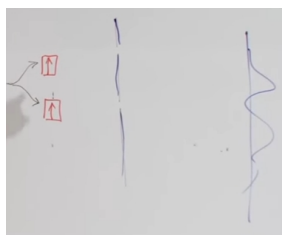
Repeat with a beam splitter

Figure 16: An experiment to observe the change in phase of a Fermion

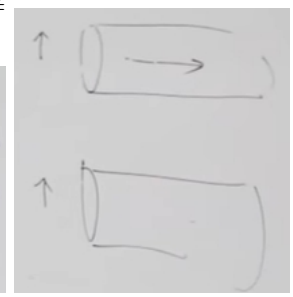
(a) Two experiments, with an without *slow* rotation



(b) Experiment with beam splitter: one electron, but waves are split. One box *only* rotates. $\psi_1(x) - \psi_2(x) \neq \psi_1(x) + \psi_2(x)$



(c) Detail of actual experiment. Magnetic field used to rotate spin, tuned so rotation is $2\pi$



# 6 Quantum Field Theory

We are entering today into the world of Quantum Field Theory. Quantum Field Theory is the description of nature as we know it, with the exception of gravity. In principle we believe that we could explain everything, except gravity, if we only had enough computational power. Of course problems tend to be too complicated. We can understand hydrogen, but boron is already too complicated, and human beings are much too complicated. Nevertheless, apart from gravity, Quantum Field Theory seems to be all there is. Today we will study second qunatization.

## 6.1 Review and Generalization of Harmonic Oscillator

The Harmonic Oscillator is the central mathematical object that goes into Quantum Field Theory. We need to understand it well. We are interested in the algebra of the operators $a^+$, $a^-$ and $N$.

Imagine many (possibly infinite number) harmonic oscillators. Think of each oscillator as an independent system of degrees of freedom, each with its own frequency.

From (23) and (24):

$$a_i^+ = \frac{P + i\omega x}{\sqrt{2\omega}}, \text{ raising or creation operator} \tag{29}$$

$$a_i^- = \frac{P - i\omega x}{\sqrt{2\omega}}, \text{ lowering or annihilation operator} \tag{30}$$

$$a^- \left|0\right\rangle = \vec{0}$$

$$N_i = a_i^+ a_i^-, \text{ number operator–this is what goes into Hamiltonian}$$

We can compute the commutators from (26) and (27), and use the fact that oscillators $i$ and $j$ are independent (if two measurements don't commute, they can not be measured simultaneously - i.e. they are not independent).

$$[a_i^+, a_j^+] = 0 \tag{31}$$
$$[a_i^-, a_j^-] = 0 \tag{32}$$
$$[a_i^-, a_i^+] = \delta_{i,j} \tag{33}$$
$$H = \hbar \sum_i \omega_i N_i \text{: the } N_i \text{ are known as } \textit{occupation numbers.}$$

We're ignoring ground state energy, as it is just an additive constant, and plays no role.

How do we label states? For one oscillator we have the occupation number. One way to specify a basis of states is to write the occupation number for each oscillator–$|n_1, n_2, n_3, ...\rangle$. One example where we need an infinite number of oscillators is an idealized violin string.

**Theorem 19** (Normalization of harmonic oscillator state–creation)**.**

$$\begin{aligned} \langle m|n\rangle &= \delta_{m,n} \\ &\Longrightarrow \\ a^+ |n\rangle &= \sqrt{n+1}\, |n+1\rangle \end{aligned} \tag{34}$$

*Proof.*

$$a^+ |n\rangle = c_n |n+1\rangle \text{, for the eigenvector } |n\rangle \text{ and some } c_n$$

Transposing and, without loss of generality, assuming $c_n$ real

$$\begin{aligned} \langle n| a^- &= c_n \langle n+1| \text{, so taking the inner product with the foregoing} \\ \langle n|a^- a^+|n\rangle &= c_n^2 \text{, from (34)} \\ &= \langle n|a^+ a^- + 1|n\rangle \text{, using (33)} \\ &= \langle n|N+1|n\rangle \text{, from (27)} \\ &= n+1 \text{, whence} \\ c_n &= \sqrt{n+1} \end{aligned}$$

□

**Theorem 20** (Normalization of harmonic oscillator state–annihilation)**.**

$$a^- |n\rangle = \sqrt{n}\, |n-1\rangle$$

*Proof.*

$$\begin{aligned}
a^- |n\rangle =& c'_n |n-1\rangle \text{ , for the eigenvector } |n\rangle \text{ and some } c_n \\
\langle n| a^+ =& c'_n \langle n-1| \text{ , wlog assume } c_n \text{ real} \\
\langle n| a^+ a^- |n\rangle =& (c'_n)^2 \langle n-1|n-1\rangle \\
\langle n| N |n\rangle =& (c'_n)^2 \text{, from (27)} \\
n =& (c'_n)^2
\end{aligned}$$

□

**Corollary 21** (Annihilation of ground state)**.**

$$a^- |0\rangle = \vec{0}$$

*Proof.* In the proof of Theorem 20, $c'_n = 0$. □

**Corollary 22** (Creation operator acting on vector)**.**

$$a_i^+ |n_1, n_2, ...n_i, ..\rangle = \sqrt{n_i + 1} |n_1, n_2, ...n_i + 1, ...\rangle$$

## 6.2 Non-relativistic Quantum Field Theory of Bosons

Particles are described by wave functions, which are functions of position, and sort of look like fields. Fields are degrees of freedom that depend on position and time, but we'll focus on position and freeze time for a while. The wave function of a particle, $\psi(x)$, the state vector in the position representation, is a function of position, but it isn't "the field". For one thing $\psi(x)$ is not Hermitian, hence it is not an observable; we observe position, momentum, angular momentum, etc, but not $\psi$.

1. $\psi$ is not an observable;
2. for multiple particles, $\psi$ is a function of all positions–$\psi(x_1, x_2)$
3. $\psi(x_1, x_2, ...x_n)$ describes the state of a *fixed number of particles.*

That is not the idea of a quantum field. We want a quantity $\Psi$ such that:

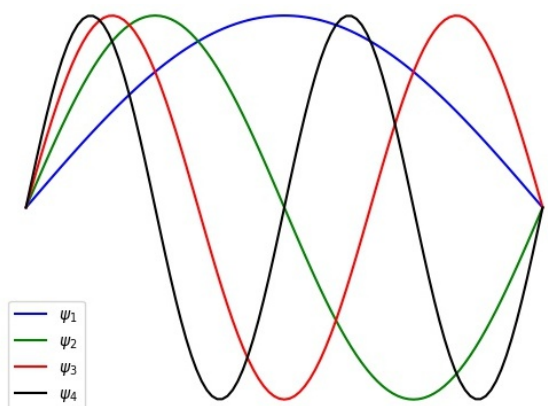
1. $\Psi$ is an observable, e.g. magnetic field, so it is an operator, not a state;
2. $\Psi$ function of one position;
3. $\Psi$ describes a system of any number of particles (even if number of particles changes).

Each field $\Psi$ corresponds to a particular type of particle: for this lecture we mean bosons.

A question was asked about using $i$ to label states. Let's go back to single particle quantum mechanics. For simplicity consider one particle in a box, whose

potential energy stops it getting out of the box. Since there's only one particles we don't need to worry about bosons versus fermions. Let's think about energy eigenstates. They are wave functions, and the lowest eigenstate $\psi_1(x)$ will be sine (lowest energy level)–Figure 17. Next $\psi_2(x)$ has one node, with higher energy, $\psi_3(x)$ two nodes,...

Figure 17: Wave function of 1-particle system in a box: $\psi_1(x)$, $\psi_2(x)$,..., using non standard notation where $i$ stands for the number of nodes plus 1. NB: this is not the same $i$ as used in (29).




Now imagine that we have many particles (bosons), some in state 1 (first energy level), some in second,..., which we will write: $|n1, n2, n3, ...\rangle$. This is identical to the way we labelled states in a multi-oscillator system (We couldn't do this for Fermions!). Because this is a complete parallel, we can *invent* operators as in (31) - (33) to *literally* create and remove particles. NB $\{1, 2, 3, ...\}$ are states, $\{n_i\}$ occupation numbers.

$$a_1^+ |n_1, n_2, ...\rangle \triangleq \sqrt{n_1 + 1}\, |n_1 + 1, n_2, ...\rangle$$

NB: this is a *definition*, chosen so we can use all the apparatus of the harmonicq oscillator.

We had a question about the physical significance of the square root. We know that

$$a^+ a^- |n\rangle = n |n\rangle \tag{35}$$

so it is reasonable to expect a square root to be associated with each operator. However, we can do better. Suppose that we don't know (35), but we did know about commutator and square root.

$$
\begin{aligned}
(a^+a^-)\left|n\right\rangle =&(a^-a^+-1)\left|n\right\rangle \\
=&\sqrt{n+1}a^-\left|n+1\right\rangle-\left|n\right\rangle \\
=&(\sqrt{n+1})^2\left|n\right\rangle-\left|n\right\rangle \\
=&(n+1)\left|n\right\rangle-\left|n\right\rangle \\
=&n\left|n\right\rangle\text{, which is (35)}
\end{aligned}
$$

NB: we are *defining* creation and annihilation operators. Don't ask *why* for a definition, *ask why it is useful*. E.g. our definition of creation is useful because we want to explore systems where the number of particles may change. It lets us create a photon, and annihilation lets us absorb particles.

**Definition 23** (Vacuum)**.** *Vacuum is the state that is annihilated by all annihilation operators:* $a^- - \left|0,0,0,....\right\rangle$.

Denote energy by $\omega_i$ for state $i$.

$$
\begin{aligned}
E =& \sum_i n_i\omega_i \\
=& \sum_i \omega_i \underbrace{a_i^+a_i^-}_{N_i}
\end{aligned}
$$

You can think of this two ways:

- oscillator energies;
- sum of all particles.

We are inventing something that is a bunch of oscillators.

- a violin string is a bunch of oscillators;
- radiation in a cavity is a bunch of oscillators.

We are building up an idea of collections of particles as harmonic oscillators; soon enough we'll see the connection with fields.

We'll consider free particles–no interactions.

Why can we ignore ground-state energy? Because constants commute with everything.

Quantum field theory is a book-keeping device for keeping track of many particles that come and go.

### 6.2.1 Digression on History of Quantum Field Theory

- Faraday invented Field Theory.

- Maxwell invented modern Field Equations (there had been wave equations before him).
- Planck introduced $\hbar$ (realized classical field theory not enough). He quantized the oscillators in cavity of black body: he didn't realize that field had to be quantized.
- Einstein realized field had to be quantized.
- Heisenberg, Pauli, Dirac put it all together around 1928.
- particle creation via harmonic oscillator? Maybe Dirac.

Quanta are the things that occupy the Occupation Numbers: if the field is the electromagnetic field we call the quanta photons.

### 6.2.2 Quantum Field Theory resumed

$\Psi(x)$ – an operator that is a function of position–one operator for each position. It acts on a space of states $\{|n_1, n_2, ...n_i, ...\rangle\}$, known as *Fock Space*–[3].

$$\begin{aligned}
\Psi(x) &\triangleq \sum_i a_i^- \psi_i(x), \text{ not Hermitean, so conjugate is} \\
\Psi^\dagger(x) &= \sum_i a_i^+ \psi_i^*(x) \\
\Psi(x) + \Psi^\dagger &\text{ is Hermitean}
\end{aligned}$$

If the $\psi$ were in the momentum representation, the coefficients would resemble Fourier coefficients. They are the quantum version of Fourier coefficients.

What if we apply $\Psi^\dagger$ to vacuum? $\Psi(x)$ annihilates it–what about $\Psi(x)^\dagger$.

$$\begin{aligned}
\sum_i |i\rangle \langle i| &= I, \text{ sum over one particle states with } \psi_i \\
\sum_i |i\rangle \langle i|X\rangle &= |X\rangle \text{ , state where particles known to be at } X\text{. But} \\
\langle i|X\rangle &= \psi^*(X), \text{ so we can write:} \\
\psi_i^*(x) |i\rangle &= |x\rangle \text{ one particle in state } i \text{ at } x\text{. But} \\
|i\rangle &= a_i^+ |0\rangle \text{ , using } a_i^+ \text{ to create the particle. Whence} \\
\psi_i^*(x) a_i^+ |0\rangle &= |x\rangle \\
\Psi^\dagger(x) |0\rangle &= |x\rangle \text{ so } \Psi^\dagger(x) \text{ creates a particle at } x.
\end{aligned}$$

- $a_i^+$ creates a particle in state $i$.
- $\Psi^\dagger(x)$ creates a particle at position $x$.

- $a_i^-$ annihilates a particle in state $i$, or gives zero.
- $\Psi(x)$ annihilates a particle at position $x$, or gives zero.

There is a separate field for each Boson.

Add vacuum to a state, we get probability of 0.5 of vacuum, 0.5 of original state. It is not the same as zero. Vacuum has length of 1!

Let's make a two particle state, at $x$ and $y$.

$$\begin{aligned}
\Psi^\dagger(y)\Psi^\dagger(x)\left|0\right\rangle &= \left|y,x\right\rangle \\
\Psi^\dagger(x)\Psi^\dagger(y)\left|0\right\rangle &= \left|x,y\right\rangle \text{, but these commute– } (31) \\
\Psi^\dagger(y)\Psi^\dagger(x) &= \Psi^\dagger(x)\Psi^\dagger(y)\text{, so} \\
\left|x,y\right\rangle &= \left|y,x\right\rangle \text{, which is the defining property of Bosons.}
\end{aligned}$$

A given quantum field describes one type of particle. There are two types of field for photons, as there are two polarizations. There is a separate field for each type of particle.

We can handle situation where number of particles is a random variable: a laser wave is a superposition of different number of photons.

# 7 Quantum Field Theory II

## 7.1 Review construction of a simple quantum field

The trick of introducing fields from particles can be done in two ways. We can start with particles and reconstruct why they can be described by fields, or start with wave fields and work out why they have quanta. Starting with particles we can describe 1 particle, 2 particles, etc, then move on to a variable number of particles. We'll start with one single type of particle. The number of photons in room changes constantly. A neutron can decay into a proton, an electron, and a neutrino. We can't rely on quantum mechanics of a single particle: we need to invent fields.

We will start with single particle quantum mechanics, a state vector $|\psi\rangle$. We can construct the inner product $\langle x|\psi\rangle = \psi(x)$ and the probability density $\psi^*(x)\psi(x)$. Normalization: $\int \psi^*(x)\psi(x)dx = 1$ gives *number of particles*. This isn't very interesting since we had only one particle to begin with.

We next introduced a collection of particle states that form an orthonormal basis. There are many ways to do this. For instance the x-states form an orthonormal basis. The eigenvectors of any Hermitian operator will form a basis. In particular the energy eigenstates of the particles in a box–Figure 17 form an orthonormal basis.

$$\int d(x)\psi_i^*(x)\psi_j(x) = \delta_{i,j} \tag{36}$$

Now we have another useful little manipulation.

**Theorem 24** (Construction of delta function from orthonormal basis)**.** *For any orthonormal basis, $\psi_i$, and any $x$ and $y$:*

$$\delta(x-y) = \sum_i \psi_i(y)\psi_i^*(x)$$

*Proof.*

$$\begin{aligned}
\sum_i |i\rangle\langle i| =& I. \text{ For any } x \text{ and } y \\
\langle y|x\rangle =& \sum_i \langle y|i\rangle\langle i|x\rangle \\
=& \sum_i \psi_i(y)\psi_i^*(x), \text{ but} \\
\langle y|x\rangle =& \delta(x-y), \text{ whence} \\
\delta(x-y) =& \sum_i \psi_i(y)\psi_i^*(x)
\end{aligned}$$

□

This is true for any set of eigenvectors of any Hermitian operator.

Now we move on to problem of studying many particles at the same time. How are we going to characterize a state? One set of states that is particularly useful is to say how many there are for each energy level. Characterize by the occupation numbers, $|n_1, n_2, ...n_i, ...\rangle$. These don't belong in the same vector space as the single particle wave functions. Do not confuse the single particle states with these more general structures. We use the same notation merely because nobody has invented anything that works better than Dirac's bras and kets.

We want to change the numbers of particles, i.e. increase and decrease occupation numbers. We introduce an oscillator for each $i$: this has nothing to do with oscillation, it is just a bookkeeping trick. We introduce creation and annihilation operators, $a_i^+$ and $a_i^-$, and make them behave the same as the oscillator creation and annihilation operators. They are often written slightly differently.

$$\begin{aligned}
a_i^+ =& a_i^\dagger \\
a_i^- =& a_i
\end{aligned}$$

We will find it convenient to introduce the field operator, which operates on Fock space.

$$\begin{aligned}
\Psi(x) \triangleq& \sum_i a_i\psi_i(x) \\
\Psi^\dagger(x) \triangleq& \sum_i a_i^\dagger\psi_i^*(x)
\end{aligned}$$

- These would be Fourier transforms of sine/cosine waves, except they are operator values, not numeric.
- $\Psi(x)$ and $\Psi^\dagger(x)$ are not Hermitian.

The following are Hermitian, and, hence, observable:

$$\Psi(x) + \Psi^\dagger(x)$$
$$\frac{\Psi(x) - \Psi^\dagger(x)}{i}$$

But right now we aren't observing them, just using them as bookkeeping devices.

We will now look at a different interpretation of $\Psi(x)$. What does it do when it operates on the vacuum, the Fock space state where all the occupation numbers are zero: $|0\rangle = |0, 0, 0, .....\rangle$? We'll write down the following formula.

$$\begin{aligned}
|x\rangle =& \sum_i |i\rangle \langle i|x\rangle \text{ resolution of the identity[2]. Now} \\
|i\rangle =& a_i^\dagger |0\rangle \text{ , i.e. a state with one particle in } i^{th} \text{ energy level, and} \\
\langle i|x\rangle =& \psi_i(x), \text{ so} \\
|x\rangle =& \sum_i \psi_i^*(x) a_i^\dagger |0\rangle \\
=& \Psi^\dagger(x) |0\rangle
\end{aligned}$$

So $\Psi^\dagger(x)$ is the operator that creates, out of the vacuum, a particle with position $x$.

- $a_i^\dagger$ creates particles with definite energy;
- $\Psi^\dagger$ creates particles at definite locations.

Dirac introduced the following terminology:

- c-number (an ordinary or classical number) commutes with everything;
- q-number (quantum operator "number") does not always commute.

## 7.2 Some illustrative calculations

We will now see what we can do with the $\Psi$ and $\Psi^\dagger$ operators.

### 7.2.1 Density of particles

The operator $\int dx \Psi^\dagger(x)\Psi(x)$ is interesting: it is not the same as $\int dx \psi^\dagger(x)\psi(x)$ (operator vs wave function). What does it do?

$$\begin{aligned}
\int dx \Psi^\dagger(x)\Psi(x) &= \int dx \sum_{i,j} a_i^+ \psi_i^*(x) a_j \psi_j(x), \text{ expanding using definition of } \Psi \text{ and } \Psi^\dagger \\
&= \sum_{i,j} a_i^+ a_j \int dx \psi_i^*(x)\psi_j(x) \\
&= \sum_{i,j} a_i^+ a_j \delta_{i,j}, \text{ from (36)} \\
&= \sum_i a_i^+ a_i. \text{ Now we use (27)} \\
&= \sum_i N_i, \text{ total number of particles.}
\end{aligned}$$

So $\int dx \Psi^\dagger(x)\Psi(x)$ is the operator representing total number of particles. Will it be finite, since there are infinitely many possible values of $i$? Unless the total energy is insanely high, the $N_i$ must fall off quickly. To make energy finite, total number of particles must be finite. $\Psi^\dagger(x)\Psi(x)$ represents particle density[6], since its integral is the total number.

Now $\Psi^\dagger(x)\Psi(x)$ is an observable, so we can measure it. To measure we integrate over a small volume, and ask how many particles are there. Like any observable it has states: you can measure and get a variety of answers.

$\{|n_1, n_2, ...n_i, ...\rangle\}$ is not the most general state: it is a basis. We can superpose to get the most general state.

### 7.2.2 Total Energy

We now determine the operator that gives the total energy. We'll assume that particles don't interact. This works for dilute particles. Even photons will interact if they are dense enough (we can't get this density in the laboratory.)

$$\begin{aligned}
E &= \sum_i N_i \omega_i \text{ sum over all modes, i.e. single particle states} \\
&= \sum_i a_i^\dagger a_i \omega_i
\end{aligned}$$

[6]The measured value is not necessarily the probability; usually will be within $\sqrt{n}$ of probability.

The $\omega_i$ from the eigenvalues of the time independent Schrödinger equation (ignoring interactions). Particles may interact with $V$, but not with each other.

$$\begin{aligned} H\psi_i =& \omega_i \psi_i \\ \big[\frac{P^2}{2m} + V(x)\big]\psi_i(x) =& \omega_i \psi_i(x), \text{ substituting} \\ P =& -i\frac{\partial}{\partial x}, \text{ we get} \\ \big[-\frac{\nabla^2}{2m} + V(x)\big]\psi_i(x) =& \omega_i \psi_i(x) \end{aligned} \tag{37}$$

We will guess the solution. Start with the observation that the expectation of the energy for one particle is given by:

$$\langle \psi | H | \psi \rangle = \int dx\, \psi^*(x)\big[-\frac{\nabla^2}{2m} + V(x)\big]\psi(x)$$

This suggests replacing $\psi$ by $\Psi$:

**Theorem 25** (Energy of multi-particle system)**.**

$$E = \int dx \Psi^\dagger(x)\big[-\frac{\nabla^2}{2m} + V(x)\big]\Psi(x)$$

*Proof.*

$$\begin{aligned} &\int dx\, \underbrace{\Psi^\dagger(x)\big[-\frac{\nabla^2}{2m} + V(x)\big]\Psi(x)}_{\text{energy density}} \\ =& \int dx \sum_{i,j} a_i^+ \psi_i^*(x)\big[-\frac{\nabla^2}{2m} + V(x)\big]\psi_j(x) a_j^- \\ =& \int dx \sum_{i,j} a_i^+ a_j^- \psi_i^*(x)\omega_j \psi_j(x), \text{ from (37)} \\ =& \sum_{i,j} a_i^+ a_j^- \delta_{i,j}\omega_j, \text{ since } \{\psi_i\} \text{ form an orthonormal basis} \\ =& \sum_i a_i^+ a_i^- \omega_i \\ =& \sum_i N_i \omega_i \end{aligned}$$

□

Notes:

- We can recover single particle QM from QFT: just go to states where total number of particles is one.

- QFT behaves like classical theory if the number of particles in the same state is very large. E.g. expectations follow laws of classical particles.
- If we don't split too many hairs, square of electromagnetic field is qualitatively related to density of photons.

You can imagine how confused everybody was before Dirac put it all together. People were confused by $\psi$: some thought it was a field, like the electromagnetic field; others thought it was just an auxiliary device for computing probabilities. If that was what is was, how was it connected with the electromagnetic field? What was the connection between particles and fields? The Schrödinger equation looked a bit like Maxwell's equations. But Maxwell's equations were not anything to do with probability amplitudes: they concerned measurable values of the field. But there was a connection, as the field was made out of quanta called photons. QFT is the resolution of that confusion.

For the electromagnetic field:

$$\begin{aligned}\vec{\Psi} =& \vec{E} + i\vec{B} \\ \vec{\Psi}^\dagger =& \vec{E} - i\vec{B} \\ \vec{\Psi}\vec{\Psi}^\dagger =& E^2 + B^2, \text{ energy density of photons/}\end{aligned}$$

Density of photons isn't a good concept, but energy density is.

$$\begin{aligned}P =& -i\frac{\partial}{\partial x} \\ \langle\psi|P|\psi\rangle =& -i\int dx\psi^*(x)\frac{\partial\psi(x)}{\partial x}\end{aligned}$$

What about the momentum operator? Do fields have momentum? A powerful laser beam will blow a hole in the wall, because it carries a lot of momentum. Fields have momentum. We will guess:

$$-i\int dx\Psi^*(x)\frac{\partial\Psi(x)}{\partial x}$$

# 8 Second Quantization

## 8.1 Digression on neutrinos

If neutrinos have mass, and they mix, do they have the same mass? No. So how do they mix and conserve energy?

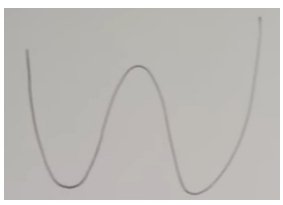
Mass is energy. The eigenstates of energy are linear superpositions of eigenstates of something else, *the type of neutrino*. The eigenstates of energy don't mix with each other, but they are mixtures themselves.

### 8.1.1 A simple example

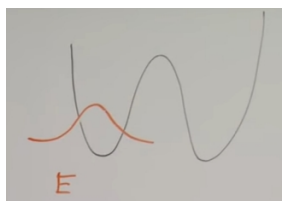
Here is simple example. Imagine a particle trapped in the potential of Figure 18a, and let's suppose for the sake of argument that the middle barrier is very high, so, to a good approximation, particle is trapped on one side or t'other (ignore tunnelling). So we can treat particle as being on the left, say. In Figure 18b we treat the barrier as a brick wall and ask what is the ground state. We might say that Figure 18b shows the ground state: it has no nodes, is as smooth as possible, and has an energy, $E$. But there must be a second ground state, from symmetry–Figure 18c, with the same energy. These are two states that have apparently the same energy–except they don't. The two wave functions in Figure 18c are not exactly eigenstates of the energy.

Figure 18: Double well potential with very high barrier

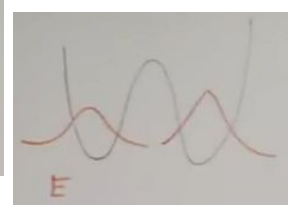
(a) The middle barrier is very high, so particle is trapped on one side.



(b) We treat the barrier as a brick wall and ask what is the ground state




(c) There must be a second ground state, from symmetry, with the same energy.




We know that for a potential like Figure 18a the eigenstates are either symmetric or antisymmetric functions.

**Theorem 26** (If $V$ is an even function, $\psi$ is either even or odd.)**.** *If $\psi$ is a continuously differentiable solution to the time independent Schrödinger equation*

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x) \text{ and} \tag{38}$$

$$\forall x V(-x) = V(x) \tag{39}$$

$$\text{then either}$$

$$\psi(x) = \psi(x), \text{ or}$$

$$\psi(x) = \psi(-x)$$

*Proof.* Define an operator $R$ (reflection in time) such that:

$$R\left|\psi\right\rangle = \left|\psi'\right\rangle, \text{ where} \tag{40}$$

$$\psi'(x) = \psi(-x)$$

(39) $\implies$ $RV = V$, and $\psi'$ satisfies (38), whence:

$$\psi'(x) = \rho\psi(x), \text{ for some constant } \rho. \tag{41}$$

WLOG we can assume that $\psi$ and $\psi'$ are both normalized, whence:

$$\begin{aligned}
|\rho| &= 1, \text{ so (40) and (41) give:} \\
R\,|\psi\rangle &= \rho\,|\psi\rangle\,. \text{ But (40) implies} \\
R^2 &= I, \text{ whence} \\
\rho^2 &= 1, \text{ i.e.} \\
&\rho \pm 1, \text{ so, in any interval where } \psi(x) \neq 0 \textit{ either} \\
\psi(x) &= \psi(-x) \textit{or} \\
\psi(x) &= -\,\psi(-x)
\end{aligned}$$

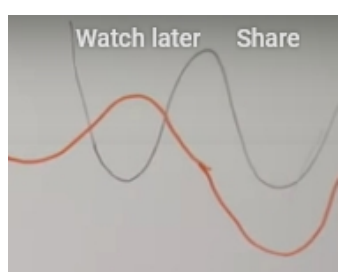
We still need to rule out the possibility that $\rho$ changes sign at a zero. Let $x_0$ be a point such that $\psi(x_0) = 0$, and, WLOG, assume that $\psi(x) = \psi'(x)$ in some interval up to $x_0-$ and $\psi(x) = -\psi'(x)$ in some interval starting $x_0+$. Since $\psi$ is continuously differentiable: then $\frac{\psi'(x_0-)}{dx} = \frac{\psi^{(}x_0-)}{dx}$. Now define $\psi'' \triangleq \big(\psi' - \psi\big)$: $\psi''$ satisfies (38), $\psi''(x_0) = 0$, and $\frac{\psi''(x_0-)}{dx} = 0$, whence $\psi'' = 0 \forall x$. $\square$

But neither LHS or RHS wave function of Figure 18c symmetric or antisymmetric. Can make symmetric or antisymmetric combinations. Figure 19a has slightly lower energy than Figure 18c: it is the true ground state[7]. Figure 19b shows first excited state: it's energy is ever so slightly larger that a wave function constrained to be on one side.

Figure 19: Wave functions for Figure 18c

(a) Wave function symmetrized(minimum in middle is slightly greater than zero-allow tunnelling)

(b) First excited state




So we mix the pure states from Figures 19a and 19b. Let's start with electron on left, and evolve.

$$\frac{\psi_L + \psi_R}{\sqrt{2}} \text{–symmetric, with energy} \qquad E_1 - \epsilon$$

$$\frac{\psi_L - \psi_R}{\sqrt{2}} \text{–antisymmetric, with energy} \qquad E_1 + \epsilon$$

[7]Remember that sign of wave function doesn't matter

What if we put a particle on the left? How does it evolve? Each is an eigenvector, so they evolve as follows:

$$\begin{aligned}\frac{\psi_L+\psi_R}{\sqrt{2}}e^{(E_1-\epsilon)t} =&\psi^+, \text{ say}\\ \frac{\psi_L-\psi_R}{\sqrt{2}}e^{(E_1+\epsilon)t} =&\psi^-\end{aligned}$$

Now what if we start with pure $\psi_L$?

$$\begin{aligned}\psi_L =&\frac{\psi^+ + \psi^-}{\sqrt{2}}, \text{ then the denominator evolves as}\\ \psi_L' =&\psi^+ e^{(E_1-\epsilon)t} + \psi^- e^{(E_1+\epsilon)t}\\ =&e^{E_1 t}\left[\psi^+ e^{-\epsilon t} + \psi^- e^{+\epsilon t}\right]\end{aligned}$$

Signs change with time, so eventually they will be opposite.

$$\begin{aligned}e^{-\epsilon t} =& -e^{\epsilon t}\\ e^{2\epsilon t} =& -1\\ 2\epsilon t =& \pi\end{aligned}$$

So $\psi_L$ becomes $\psi_R$!

### 8.1.2 Neutrinos

The phenomenon of mixing goes with the phenomenon of oscillation. Neutrinos behave like that:

- the analog of the left wave function is the neutrino that is made when a particle decays to an electron plus a neutrino;
- the analog of the right wave function is the neutrino that is made when a particle decays to a muon plus a neutrino.

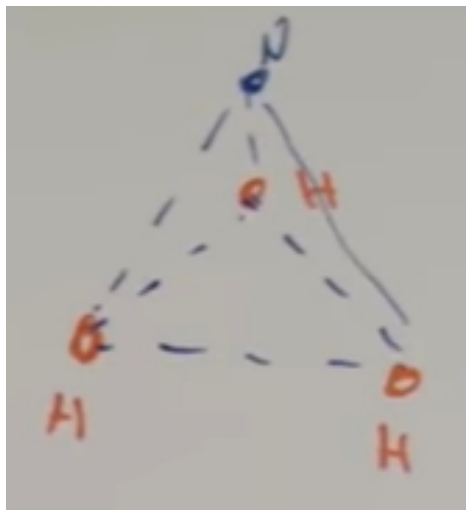
The real eigenstate of the energy is a linear combination of the muon and electron neutrino. When a neutrino is made, it is either in a state of an electron neutrino or a muon neutrino. If you wait a while it will change to the other state. –electron neutrino changes to muon neutrino. Now the neutrino can be scattered, and produce an electron (from electron neutrino) or a muon. But wait a while and see what is produced: electron neutrino becomes muon neutrino, and can produce muons. This type of neutrino will oscillate! If you have a very long baseline you'll see the oscillations

What controls size of $\epsilon$? All sorts of details, mass of particle, height of barrier, width of barrier. It is extremely sensitive.

The neutrino masses are the energy levels. They start small, but mixing splits the energy a bit. There is no barrier; there is a coupling in the Hamiltonian which, for whatever reason takes an electron neutrino to a muon neutrino and vice versa. In Figure 18c the coupling is due to tunnelling.

Ammonia has a tetrahedral shape–Figure 20. The Nitrogen likes to sit a certain distance of the base of the triangle (energy lowest). But there is another state which is symmetric, with Nitrogen below: $N$ tunnels through $H_3$ plane.

Figure 20: Ammonia has a tetrahedral shape

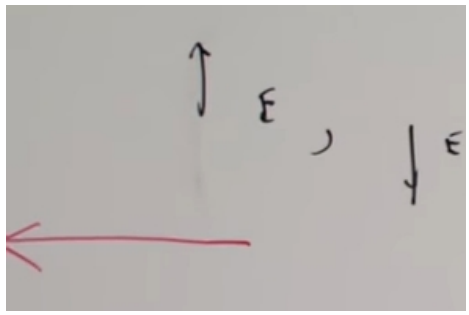



We've also seen this with spins. Without any magnetic field up and down have the same energy. Now apply a weak magnetic field–Figure 21. What are the eigenstates? Left, and right. But these are just linear combinations of up and down.

$$|l\rangle = \frac{1}{\sqrt{2}}\big[\,|u\rangle + |v\rangle\,\big]$$
$$|r\rangle = \frac{1}{\sqrt{2}}\big[\,|u\rangle - |v\rangle\,\big]$$

The eigenstates have slightly different energy because of the magnetic field: we have the same situation with slightly different energy levels. What happens if we start the electron up? It precesses, moving between down and up. There is no tunnelling, just a term in the Hamiltonian.

Figure 21: Spins in a weak magnetic field



If there is something that an up electron can do, and down can't, and vice version, we can create the electron as an up, and later do an experiment to see whether it is behaving like a down.

The strength of field only controls speed of precession. The analogue of the magnetic field is the mixing of the neutrinos: this is a very small parameter, and we don't know where it came from. It doesn't fit into the Standard Model.

Neutrino oscillation: the energy is also affected by spin.

This is believed to be the explanation of the deficit of solar neutrinos: by the time they get here, some have transformed.

### 8.1.3 Scientific American Article on whether electron is a sphere

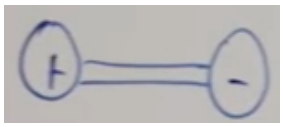
What would it mean for the electron to be a sphere?[4]

What was measured was the electric dipole moment, the charge times the distance between them, and it represents an off centred distribution. The centre of the dumbbell is the centre. For a charge distribution not to have a dipole does not mean that it is a sphere. A charge distribution can have many multipole moments. What kind of shape corresponds to a dipole? One with an imbalance of charge. We could have a distribution shaped like an oblate spheroid, or any kind of ellipsoid. This would not have a dipole moment, because it is uniformly distributed about any plane passing through centre. It might have a quadrapole moment, or an octopole moment.

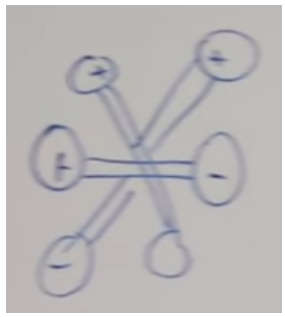
In quantum mechanics there is something odd about the notion of spherical symmetry. Figure 22a depicts a quantum dumbbell.

Figure 22: Quantum Dumbbells and electric dipoles

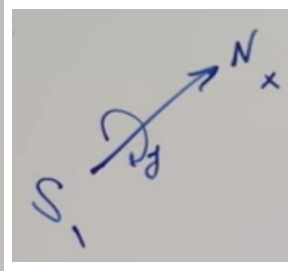
(a) Quantum Dumbbell - e.g. a molecule




(b) Quantum Superposition



(c) Spinning electron, showing hypothetical imbalance of charge.




Let's take the ground state of the molecule, which is described by its angular momentum about the centre (ignore vibrations). If molecule is a boson, the ground state will have angular momentum 0: wave function is completely symmetric for all rotations. So, independent of shape of molecule, the ground state is rotationally symmetric. Maybe it is a superposition of states, such as Figure 22b, so wave function is symmetrized. Does Figure 22b have a dipole moment? Answer is ambiguous, depending on the experiment.

- Take a sudden quick snapshot; we'd discover it was lopsided with dipole moment. How fast does camera have to be? What resolution do we need? Snapshot measures angle; system would be in a state of definite orientation, so can't measure angular momentum (Heisenberg).

- What is energy of first excited state? What if it is enormous? More than we have in our photons. Then apparatus cannot resolve angle. We measure average: a sphere. Mesons are also (smaller) dumbbells. Kicking it into first excited state is huge compared to ordinary photons. All we can measure shows it is a sphere on average.

Question of whether it a sphere or not is confusing.

Bosons can have spin zero, but electrons can't. They have an axis–Figure 22c. Given that direction, is there a charge imbalance?

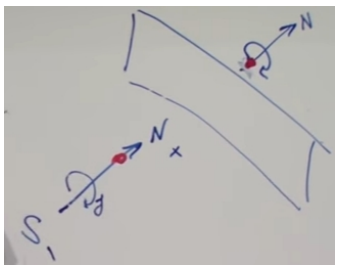
What did they measure? Once you know that electron has an axis, we can ask whether there is an imbalance (small compared to overall negative charge–e.g. move charge from centre). Is there an imbalance as shown in Figure 22c? Is there a correlation between electrical charge displacement and the magnetic direction? This is what would be called the electric dipole moment of the electron.

Some symmetries forbid electric dipole moment, e.g. reflection symmetry. If the mirror image of an electron is another electron. Figure 23a shows that reflection doesn't preserve relationship between dipole and field direction. So,

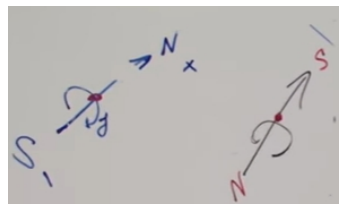
*if the laws of physics don't distinguish between left hand and right hand*, there is no dipole moment.

Figure 23: Symmetries and dipoles

(a) Mirror image of an electron. Magnetic field has same directions, since it comes from spinning charge, but charge imbalance is reversed.




(b) Mirror image of an electron. Magnetic field has same directions, since it comes from spinning charge, but charge imbalance is reversed.




But the laws of physics do distinguish between left hand and right hand (e.g. neutrinos). But there another more accurate symmetry, time reversal. Figure 23b shows the direction of spin being reversed, hence the magnetic field reversed, but the charge distribution is unchanged. So time reversal would preclude a dipole, if it is a good symmetry of nature.

But time reversal is not a good symmetry of nature, either.

Standard Model predicts a very small dipole moment that is too small to detect. Experiment has failed to detect dipole moment.

## 8.2 Second Quantization of Bosons

How are fields connected with Fourier transforms? In ordinary QM we have wave functions; in the position representation, and the squares define probabilities of the particle being at a particular point. We can Fourier transform the wave functions, and the transform is the representation in momentum space.

$$\psi(x) \rightarrow \psi^*(x)\psi(x) = P(x), \text{ probability in position representation}$$
$$\widetilde{\psi}(p) \rightarrow \widetilde{\psi}^*(p)\widetilde{\psi}(p) = P(p), \text{ probability in momentum representation, where}$$
$$\widetilde{\psi}(p) = \int \frac{dx}{\sqrt{2\pi}} \psi(x) e^{-ipx}$$
$$\psi(x) = \int \frac{dp}{\sqrt{2\pi}} \widetilde{\psi}(p) e^{+ipx}$$

Now let's go to field theory, with creation and annihilation operators.

$$\Psi(x) = \sum_i a_i^- \psi_i(x)$$

$$\psi_i(x) = e^{ipx} \text{ for a free particle (different } i\text{!), so}$$

$$\Psi(x) = \int \underbrace{\frac{dp}{\sqrt{2\pi}}}_{\text{by convention}} a^-(p) e^{ipx}, \text{ where annihilation operator}$$

The momentum states are now represented by a continuous parameter, $p$, not a discrete $i$. Moreover $a^-(p)$ removes 1 particle with momentum $p$: it plays same role as $\widetilde{\psi}(p)$. If there is no particle to remove, $a^-(p)$ annihilates the state.

$$\Psi^\dagger(x) = \int \frac{dp}{\sqrt{2\pi}} a^+(p) e^{ipx} \text{ creates particle at position } x$$

There is a similarity between wave functions in position and momentum space and creation and annihilation operators for particles and given position or momentum. And we can invert the equations.

$$a^-(p) = \int \frac{dx}{\sqrt{2\pi}} \Psi(x) e^{-ipx}$$
$$a^+(p) = \int \frac{dx}{\sqrt{2\pi}} \Psi^\dagger(x) e^{ipx}$$

Remember there is a field operator for each kind of particle.

If we recall the commutators:

$$[a_i^+, a_j^-] = \delta_{i,j}$$
$$[a_i^+, a_j^+] = 0$$
$$[a_i^-, a_j^-] =$$

We can prove the following.

$$[\Psi^+(x), \Psi^-(y)] = \delta(x-y), \text{ introduces Heisenberg uncertainty!}$$
$$[\Psi^+(x), \Psi^+(y)] = 0$$
$$[\Psi^-(x), \Psi^-(y)] = 0$$
$$[\Psi_R^+(x), \Psi_I^-(y)] = \delta(x-y), \text{ can not measure simultaneously at same point!}$$

Makes sense in relativity: if measuring field at one point kicked value at a distant point, it would violate causality/locality. If there is a spacelike separation between measurements one cannot affect another!

Fermions don't work like this! One measurement can kick another arbitrarily far away. This sounds crazy–or there is no way to measure a Fermion Field.

# 9 Quantum Field Hamiltonian

## 9.1 Particle Field Interactions

In the previous lectures we talked about Quantum Fields, the quanta of those fields, the creation and annihilation operators, and examined the Hamiltonian for a very simple field–non-interacting particles satisfying Schrödinger's equation. The second term counts particles, giving each energy $V(x)$.[8]

$$H = \int dx \big[\Psi^\dagger(x)\big[\frac{-\nabla^2}{2m}\Psi(x)\big] + V(x)\Psi^\dagger(x)\Psi(x)\big]$$

$\Psi^\dagger(x)\Psi(x)$ represents the density of particles at $x$, so $\Psi^\dagger(x)\Psi(x)V(x)$ counts particles at $x$, and gives each one energy $V(x)$.

We restrict to the special case of constant potential energy: each particle has energy $mc^2$.

$$H = \int dx \big[\Psi^\dagger(x)\big(\frac{-\nabla^2}{2m}\Psi(x)\big) + mc^2\Psi^\dagger(x)\Psi(x)\big] \tag{42}$$

How do we see from (42) that momentum is conserved? The role of the Hamiltonian is to update state, $\phi$, say.

$$\begin{aligned} |\phi(t+\epsilon)\rangle =& (1 - i\epsilon H)\,|\phi(t)\rangle \text{ State} \\ =& |\phi(t)\rangle - i\epsilon H\,|\phi(t)\rangle \end{aligned} \tag{43}$$

What would it mean to say momentum is conserved? It means the Hamiltonian, when it acts on a state with definite momentum, doesn't change momentum. Whatever the Hamiltonian does, if it acts on a state with given momentum, it must give back the same momentum, *if momentum is conserved.* It might change other things, but not the momentum. Let's see how this works with (42). Rewrite 2nd term using momentum variables (Fourier transform).

$$\begin{aligned} \Psi(x) &= \int \frac{dp}{\sqrt{2\pi}} \underbrace{\widetilde{\Psi}(p)}_{\substack{\text{annihilation}\\ \text{operator}}} e^{ipx} \\ \Psi^\dagger(x) &= \int \frac{dp}{\sqrt{2\pi}} \underbrace{\widetilde{\Psi}(p)}_{\substack{\text{creation}\\ \text{operator}}} e^{-ipx} \end{aligned} \tag{44}$$

[8] We are not restricted to 3D: x may be a vector.

The factor is actually $\left(\frac{1}{\sqrt{2\pi}}\right)^n$ where $n$ is number of dimensions.

$$\begin{aligned}\int dx\big(mc^2\big)\Psi^\dagger(x)\Psi(x) =& \frac{mc^2}{2\pi}\int \widetilde{\Psi^\dagger}(p_2)\widetilde{\Psi}(p_1)e^{i(p_1-p_2)x}dxdp_2dp_1\\ =& \frac{mc^2}{2\pi}\int \widetilde{\Psi^\dagger}(p_2)\widetilde{\Psi}(p_1)\delta(p_1-p_2)dp_2dp_1\\ =& \frac{mc^2}{2\pi}\int \widetilde{\Psi^\dagger}(p)\widetilde{\Psi}(p)dp\end{aligned}$$

This just removes a particle with momentum $p$ and puts it back with same momentum; momentum isn't changed. The key thing was the integration over *space*, which introduced the $\delta$ function for momentum. Before we repeat with the 1st term of (42) we will see that this can be more general. Let's suppose we had a term in the Hamiltonian for a whole bunch of fields: $\Psi^\dagger(x)\Psi(x) \to \Psi_A^\dagger(x)\Psi_A(x) + \Psi_B^\dagger(x)\Psi_B(x) + ....$ We would end up with an integral:

$$\begin{aligned}&\int dp_{1A}dp_{1B}...dp_{2A}dp2B\Pi\Psi(p_1)\Pi\Psi(p_2)e^{i(\sum p_1-\sum p_2)x}dx\\ =&\int dp_{1A}dp_{1B}...dp_{2A}dp_{2B}\Pi\Psi(p_1)\Pi\Psi(p_2)\delta(\sum p_1-\sum p_2)\end{aligned}$$

The first sum is the momentum we put in, the second the momentum that we take out. Total that we take out = total that we put in. Whenever we see a Hamiltonian of this form we have sum of outgoing momentum = sum incoming.

What is really going on is the fact that the Hamiltonian is translation invariant. Notice that the integral over space that gives rise to the delta functions, and to momentum conservation. Notice that the momentum of each type of particle isn;t necessarily conserved, but the total momentum is.

Let's look at derivatives. If we calculate $\nabla^2\Psi^\dagger(x)$, only one part of (44) is affected, $e^{-ipx}$, since the remainder is doesn't involve $x$.

$$\begin{aligned}-\nabla^2\Psi^\dagger(x) &= \int \frac{dp}{\sqrt{2\pi}}p^2\widetilde{\Psi^\dagger}(p)e^{-ipx}, \text{ so}\\ \int dx\Psi^\dagger(x)\frac{-\nabla^2}{2m}\Psi(x) &= \int \frac{p^2}{2m}\widetilde{\Psi^\dagger}(p)\widetilde{\Psi}(p)dp \qquad (45)\end{aligned}$$

This adds the kinetic energy of all the particles.

Note: the left hand side of (45) looks negative, but it is meant to be kinetic energy, which is positive. Integrating by parts we get:

$$\int dx\Psi^\dagger(x)\frac{-\nabla^2}{2m}\Psi(x) = \frac{1}{2m}\int dx\frac{\partial}{\partial x}\Psi^\dagger(x)\frac{\partial}{\partial x}\Psi(x)$$

This looks like any other Hamiltonian, where you see squares of derivatives of the fields.

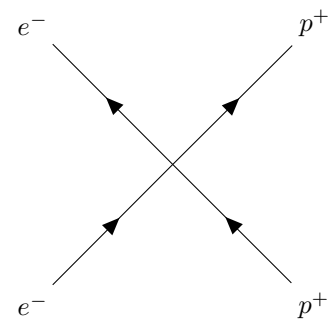
Imagine two species of particles: fake electrons and protons that happen to be bosons.[9]

$$H = \int dx \Psi_e^\dagger(x) \frac{-\nabla^2}{2m_e} \Psi_e(x) + \int dx \Psi_p^\dagger(x) \frac{-\nabla^2}{2m_p} \Psi_p(x) + g \int dx \underbrace{\Psi_e^\dagger(x)\Psi_p^\dagger(x)\Psi_e(x)\Psi_p(x)}_{\text{Figure 24}}$$

where $g$ is a coupling constant–proportional to probability of interaction.

The integrand in last term annihilates a proton and an electron if it finds them at point $x$, then creates an electron and a proton.

Figure 24: $\Psi_e^\dagger(x)\Psi_e^\dagger(x)\Psi_e(x)\Psi_p(x)$: annihilate and electron and proton if they are at the same place, then create an electron and a proton at the same place–i.e. scatter them.



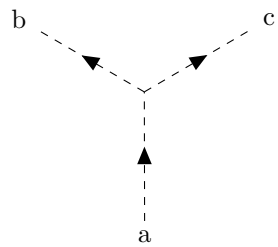

- Total momentum is conserved, not that of individual particles.
- Real electrons and protons interact through the Coulomb potential, but this has been ignored.
- If experiment shows that particles interact, then there must be a term in the Hamiltonian.
- This is the sort of term for a short range interaction.

Imagine we find particle decay in an experiment, as shown in Figure 25a. We might try a term: $\Psi_b^\dagger(x)\Psi_c^\dagger(x)\Psi_a(x)$. Since space it isotropic, the interaction can happen anywhere: $\int \Psi_b^\dagger(x)\Psi_c^\dagger(x)\Psi_a(x)dx$, or $g\int \Psi_b^\dagger(x)\Psi_c^\dagger(x)\Psi_a(x)dx$, where we have introduced the *coupling constant*, $g$. We can show that momentum is conserved.

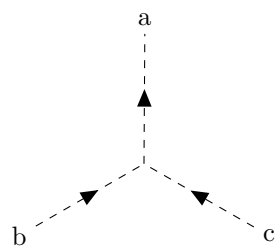
[9] We aren't quite ready for Fermions.

Figure 25: $\Psi_b^\dagger(x)\Psi_c^\dagger(x)\Psi_a(x)dx + \Psi_a^\dagger(x)\Psi_c(x)\Psi_b(x)$

(a) Particle $a$ decays to $b$ and $c$: $\Psi_b^\dagger(x)\Psi_c^\dagger(x)\Psi_a(x)$




(b) We need the reverse of (a) to make the Hamiltonian be Hermitian: $\Psi_a^\dagger(x)\Psi_c(x)\Psi_b(x)$




Now what goes in the Hamiltonian must be Hermitian, so the evolution is unitary. This means we need:

$$g\int \Psi_b^\dagger(x)\Psi_c^\dagger(x)\Psi_a(x)dx + g\int \Psi_a^\dagger(x)\Psi_c(x)\Psi_b(x)$$

which reverses the process of Figure 25a: $b + c \to a$–Figure 25b.

We now have a the basic rules of a simple version of quantum field theory:

- creation and annihilation operators;
- fields made out of creation and annihilation operators;
- fields are functions of position;
- the fields can be thought of as creation and annihilation operators at a definite position;
- each type of particle has its own field;
- we write down the Hamiltonian with kinetic energies of particles and other concoctions.

In the first round of QFT the concoctions largely came from experiment. A classic example it beta decay: a neutron decaying into a proton, an electron, and a neutrino–$g\int \Psi_\nu^\dagger(x)\Psi_e^\dagger(x)\Psi_p^\dagger(x)\Psi_n(x)dx$ (need to make Hermitian).

Let's go one step further. Imagine that we made $\epsilon$ a little larger in (43):

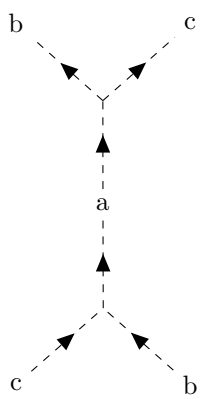
$$|\phi(t+\epsilon)\rangle = |\phi(t)\rangle - i\epsilon H\,|\phi(t)\rangle - \frac{\epsilon^2}{2}H^2\,|\phi(t)\rangle$$

What is the effect of $H^2$? We can multiply it out, but the easy way is to use the pictures. E.g., Figure 25a becomes a scatter: $b + c \to a \to b + c$. Or Figure 26a shows a more complex situation. You can have a forest of interactions that

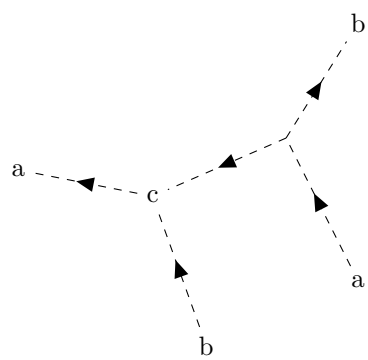
are mediated by a single term in the Hamiltonian. This is why Quantum Field Theory is powerful: a small number of terms in the Hamiltonian can lead to a huge number of interactions, all with the same coupling constant.

Figure 26: What is the effect of $H^2$?

(a) $\Psi_c^\dagger \Psi_b^\dagger \Psi_a \Psi_a^\dagger \Psi_b \Psi_c$: annihilate $c$ and $b$ to create an $a$, then annihilate the $a$ and create a $b$ and $c$: we have created a scattering, and the total momentum should remain the same.



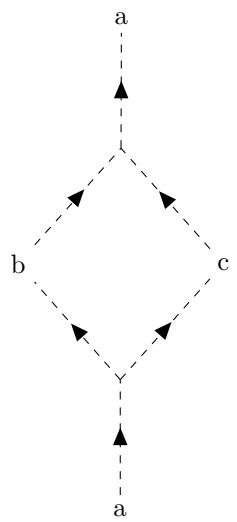

(b) Another possibility is for $a$ to come in, be annihilated producing a $b$ and a $c$. Suppose there is another $b$ around: if it can find another $c$ to annihilate with, it can produce a $a$. So iterating the Hamiltonian we also have a process where $a$ and $b$ scatter by exchanging a $c$: this is called an exchange diagram.




Figures 26a and 26b have a coupling constant $g^2$, since there are two vertices.

What about $a \to (b+c) \to a$? Fugure 27 is a self energy diagram for $a$. A real $a$ has a probability of being a $b+c$, and it behave like a $b+c$ close together some times.

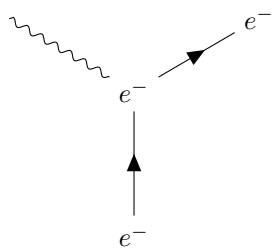
Figure 27: Self energy diagram for $a$.




One example of this would be an electron emitting a photon–Figure 28 shows a self energy diagram for the electron.

Figure 28: Scatter electron and emit photon(field $A$):$A\Psi_e^+\Psi_e^-$. Sometimes what you thought was an electron behaves like an electron + photon. Don't take it too literally: this happens, then this, then that. The pictures just come out of iterating the Hamiltonian twice. If we look, we will screw up electron! If you scatter another electron against this electron, you will find some of the time, you will see that what you thought was an electron is an electron + photon. The probability to find photon is small, about 1%. It should be thought of as a correction to the structure of an electron, not a new process.

(a) Electron comes in and emits photon–$A\Psi_e^+\Psi_e^-$




(b) One other thing that can happen: emit photon and reabsorb. The photon is the electric field of the electron. If electron is moving at the speed of light it behaves like a superposition of an electron and an electron+photon.

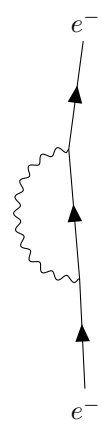



Figure 28 doesn't depict what is literally happening; they are shorthands for the mathematics of squaring the Hamiltonian. What is actually happening is a superposition. Energy isn't just the sum of the particles: it includes these interactions.

We have just seen the basic physics: the rest is technology and tricks to calculate things.

## 9.2 The Dirac Equation

Before we start on Fermion *fields* we will start with the Dirac equation, which uses relativity theory: we will study the relativistic electron. We will start with an electron that moves in one direction along a line: this isn't a realistic electron, since electrons move in 3 dimensions, but it is the easiest place to get started. Why did Dirac want to handle the relativistic case? In a hydrogen atom electrons move at an appreciable proportion of the speed of light, about 1% of the speed of light, so relativity is a significant correction. In a heavy

atom, electrons are closer to the nucleus and can be moving with even more appreciable velocities.

In order to build a Schrödinger equation for the electron, we start with the classical relation for energy and momentum.

$$
\begin{aligned}
E =& \frac{p^2}{2m}. \text{ In order to go from classical to quantum we need operators} \\
P =& -i\frac{\partial}{\partial x} \\
H =& i\frac{\partial}{\partial t}, \text{ and we'd get the Schrödinger equation. Relativistically} \\
E^2 =& p^2 + m^2, \text{ setting } c^2 = 1. \text{ This gives:} \\
-\frac{\partial^2}{\partial t^2}\phi(x) =& -\frac{\partial^2}{\partial x^2}\phi(x) + m^2\phi(x)
\end{aligned}
$$

This was already known; it is the Klein-Gordon equation. But Dirac wanted this:

$$
\begin{aligned}
i\frac{\partial\psi}{\partial t} =& H\psi, \text{ to suit his version of quantum mechanics.} \\
=& \Big(\sqrt{-\frac{\partial^2}{\partial x^2} + m^2}\Big)\psi
\end{aligned}
$$

What could this possibly mean? What is the operator on the right hand side? It looks ugly. Dirac decided to start again. Imagine a particle moving along the x axis with the speed of light (massless).

$$
\begin{aligned}
cP =& E, \text{ moving to right. Setting } c^2 = 1 \\
i\frac{\partial\psi}{\partial t} =& -i\frac{\partial\psi}{\partial x}, \text{ or} \\
\frac{\partial\psi}{\partial t} + \frac{\partial\psi}{\partial x} =& 0
\end{aligned}
\tag{46}
$$

Any function of x-t will satisfy this equation: $\psi(x,t) = \psi(x-t)$. It describes a wave, moving to the right with the speed of light. But three things are wrong with this equation.

1. It allows particles with negative momentum, which have negative energy. This sounds very dangerous. We can keep filling up the world with particles of negative energy, lowering the total energy without bound.

2. It only has particles that move to the right; we can imagine a theory in which particles only move to the right, but it wouldn't look like our world.

3. The electron has no mass, so it moves with the speed of light.

Dirac solved problem 2 by imagining two species of electrons, one moving to the right, the other to the left, labeled $\psi_1$ and $\psi_2$. A particle can be one species or the other, *or a superposition*.

$$\begin{aligned}\frac{\partial\psi_1}{\partial t}+\frac{\partial\psi_1}{\partial x}=&0, \text{ moving to right}\\ \frac{\partial\psi_2}{\partial t}-\frac{\partial\psi_2}{\partial x}=&0, \text{ moving to left}\end{aligned}$$

Dirac then imagined a particle with two degrees of freedom, $x$ and $\{1,2\}$. Make the second degree into a matrix index, and introduce an observable, $\alpha$, to tell us whether particle is 1 or 2 (a bit like a spin). We think of the $\psi$s as a column vector, and introduce a matrix whose eigenvectors are $\psi_1$ or $\psi_2$. Here is such a matrix (observable).

$$\underbrace{\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}}_{\text{Dirac's }\alpha}\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix}$$

Then he wrote:[10]

$$\begin{aligned}H=&\alpha P\\ i\frac{\partial\psi}{\partial t}=&-i\alpha\frac{\partial\psi}{\partial x}, \text{ matrix equation!}\end{aligned}\tag{47}$$

We have fixed problem 2, particles moving to the right only. To fix problem 3, absence of mass, Dirac added a term to give the electron mass.

$$\begin{aligned}H=&\alpha P+\beta m, \text{ where } \beta \text{ is some matrix. Now we want}\\ E^2=&P^2+m^2\\ H^2=&\big(\alpha P+\beta m\big)\big(\alpha P+\beta m\big)\\ =&\underbrace{\cancel{\alpha^2}}_{\substack{\text{we know}\\ \alpha^2=1}}P^2+\underbrace{\cancel{\beta^2}}_{\substack{\text{we need}\\ \beta^2=1}}m^2+\underbrace{\cancel{(\alpha\beta+\beta\alpha)}}_{\substack{\text{we need}\\ \alpha\beta+\beta\alpha=0}}Pm, \text{ then}\\ =&P^2+m^2, \text{ as desired.}\end{aligned}\tag{48}$$

We know that the Pauli matrices anticommute, and we know:

$$\begin{aligned}\alpha=&\sigma_3, \text{ so choose}\\ \beta=&\sigma_2.\end{aligned}$$

[10] LS said that Dirac didn't actually do this: he plunged right in and did it in 3 dimensions. Or maybe he did the 1 dimensional case first, and didn't write it up.

We have the 1-dimensional version of Dirac's equation.

$$
\begin{aligned}
i\frac{\partial\psi}{\partial t} =& ap\psi + \beta m\psi, \text{ where } \psi \text{ is a two component spinor.} \\
i\frac{\partial}{\partial t}\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} =& -i\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} + \begin{pmatrix}0 & m\\ m & 0\end{pmatrix}\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix}
\end{aligned}
$$

So

$$
\begin{aligned}
i\frac{\partial\psi_1}{\partial t} + i\frac{\partial\psi_1}{\partial x} =& m\psi_2 \\
i\frac{\partial\psi_2}{\partial t} - i\frac{\partial\psi_2}{\partial x} =& m\psi_1
\end{aligned}
$$

So the mass term couples the two equations, and makes the particle move slower then light. LS suspects that Dirac knew that mass is coupled to ability to move left or right. A Lorentz transformation will not convert a massless particle moving to the left to one moving right, or vice versa. A massive particle moves with a speed less that $c$, and Lorentz transformation can convert it to one moving in the opposite direction.

We still have two problems, which we will deal with in the next section:

1. We still have negative energy particles, left moving and right moving.
2. We need a 3 dimensional equation.

# 10 Fermions and the Dirac equation

## 10.1 Second Quantization of Fermions

We will generalize what we said earlier about second quantization. We made the fields that govern multiple particle systems out of creation and annihilation operators. What is the difference between fermions and bosons?

If you have two bosons, you can put them into the same state, and the creation and annihilation operators commute. There is nothing strange about commuting, but anticommuting is odd. Suppose we have two particles, and we want to say that when we interchange them the wave function or state vector changes sign. Let $|x, y\rangle$ represent a particle in slots 1, with position $x$ and one in slots 2 with position $y$, so we have two 3 dimensional positions. $\Psi^\dagger(x)\Psi^\dagger(y)\,|0\rangle$ will create. such a pair.

For bosons, operators commute and:

$$
\begin{aligned}
|x, y\rangle =& |y, x\rangle \text{ , because} \\
\Psi^\dagger(x)\Psi^\dagger(y) =& \Psi^\dagger(y)\Psi^\dagger(x), \text{ because} \\
[\Psi^\dagger(x), \Psi^\dagger(y)] =& 0
\end{aligned}
$$

For Fermions:

$$\begin{aligned}
|x,y\rangle =& -|y,x\rangle \text{ , so we need} \\
\Psi^\dagger(x)\Psi^\dagger(y) =& -\Psi^\dagger(y)\Psi^\dagger(x), \text{ or} \\
\Psi^\dagger(x)\Psi^\dagger(y)+\Psi^\dagger(y)\Psi^\dagger(x) =& 0. \text{ This is usually written:} \\
\{\Psi^\dagger(x),\Psi^\dagger(y)\} =& 0, \text{ where} \\
\{A,B\} \triangleq& AB+BA, \text{ the anticommutator.}
\end{aligned}$$

In general we can go from Bosons to Fermions by replacing the commutator with the anticommutator.

NB $\Psi^\dagger(x)\Psi^\dagger(x) = \frac{1}{2}\{\Psi^\dagger(x)\Psi^\dagger(x)\} = 0$, so we can't have two identical particles at same position.

- For Bosons there is a clear distinction between creation and annihilation operators, since $[a,a^\dagger] = 1$–Equation (33), and $[a^\dagger,a] = -[a,a^\dagger] = -1$. We can't confuse them because the spectrum of energies from 0 to $\infty$ can't be turned upside down. Creation operators never give 0!.
- For Fermions $\{a^\dagger,a\} = \{a,a^\dagger\} = +1$: from the algebra there is no difference between creation and annihilation operators. They do different things, so they are different, but we can't see this from the algebra alone.

Consider occupation numbers of Fermion with only one state: creation operator takes 0 to 1; if we create again the operator annihilates, leaving 0. There is complete symmetry between empty and filled, between creation and annihilation (you can exchange them mathematically: it doesn't mean they are the same; they have difference physical meanings).

$$\begin{aligned}
a^2 =& 0 \\
(a^\dagger)^2 =& 0
\end{aligned}$$

## 10.2 The Dirac Equation in 3 dimensions

### 10.2.1 Review of 1D Dirac Electron

We have already figured out how to deal with the fact that the simplest equation for an electron doesn't have a mass– (46). If $\psi$ is a wave function for a massless electron.

$$\begin{aligned}
i\frac{\partial\psi}{\partial t} =& -i\frac{\partial\psi}{\partial x}, \text{ from} \\
H\Psi =& P\Psi, \text{ a massless relativistic particle}
\end{aligned}$$

This was a failure as a theory of 1 dimensional electrons. We introduced a 2nd species that propagates to the left.

$$\begin{aligned} i\frac{\partial \psi_1}{\partial t} =& -i\frac{\partial \psi_1}{\partial x} \\ H\Psi_1 =& P\Psi_1 \\ i\frac{\partial \psi_2}{\partial t} =& i\frac{\partial \psi_2}{\partial x} \\ H\Psi_2 =& -P\Psi_2 \end{aligned}$$

We combined into these two wave functions into a column vector, adding a new degree of freedom, handedness, $\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$. One component can have positive or negative energy, and it moves to the right; the other component can have positive or negative energy, and it moves to the left. We invented a matrix, $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, with $\psi_1$ and $\psi_2$ as eigenvectors, so we had (47).

$$H\psi = \alpha P \psi$$

We have both wave functions in the same structure, at the cost of doubling the number of components of the field. That didn't get us a mass, but it got stuff moving in both directions. We then invented another matrix $\beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and in (48) wrote $H = \alpha P + \beta m$. Note[11] $\alpha^2 = \beta^2 = 1$, and $\alpha\beta + \beta\alpha = 1$, whence $H^2 = P^2 + m^2$. We combined left moving and right moving, then added coupling to give mass. It is in this sense that a mass term switches you from left moving to right moving[12].

How do we do this in 3D?

### 10.2.2 The Dirac Equation in 3 dimensions

Let's go back to the massless case. We drop $m$ temporarily. Next momentum is a 3 vector, $\vec{P}$, so we want $\alpha\vec{P}$ to be invariant under rotation. We make $\alpha$ a vector, $\vec{\alpha}$, so we can dot product with $\vec{P}$.

$$\begin{aligned} H =& \vec{\alpha} \cdot \vec{P} \\ =& \alpha_x P_x + \alpha_y P_y + \alpha_z P_z \end{aligned}$$

[11] The fact that these are Pauli matrices is just incidental at this point.

[12] It doesn't really, but you can think of it that way.

What do the $\alpha_i$ have to do to make this sensible in the massless equation? We want:

$$\begin{aligned}
H^2 =& P_x^2 + P_y^2 + P_z^2, \text{ so we need} \\
\alpha_x^2 =& 1 \\
\alpha_y^2 =& 1 \\
\alpha_z^2 =& 1 \\
\alpha_y\alpha_x + \alpha_x\alpha_y =& 0 \\
\alpha_z\alpha_x + \alpha_x\alpha_z =& 0 \\
\alpha_x\alpha_y + \alpha_y\alpha_z =& 0, \text{ which can be written:} \\
\{\alpha_y, \alpha_x\} =& 0 \\
\{\alpha_z, \alpha_x\} =& 0 \\
\{\alpha_x, \alpha_y\} =& 0
\end{aligned}$$

Dirac needed three matrices that satisfy these equations; the three Pauli matrices do exactly that:

$$\begin{aligned}
\vec{\alpha} =& \vec{\sigma}, \text{ so} \\
H =& \vec{\sigma} \cdot \vec{P}
\end{aligned}$$

You can think of this as a particles that has spin and momentum. Energy is the component of spin along the direction of motion. The spin is always $\pm 1$, so it looks as if we have two cases, where spin is either in the same direction as momentum, or the opposite; again we have a massless particle with positive and negative energy, and whose spin is locked so its axis is along the momentum vector.

Spin "drops into your lap" for free in the Dirac equation!

Dirac's logic, which was not necessarily unimpeachable, was that he wanted to write an equation for the Hamiltonian in terms of the momentum without having an insane square root. He wrote $H = \vec{\alpha} \cdot \vec{P}$ because he wanted it to be linear in the momentum. He wondered what $\alpha$ was, because it couldn't just be a bunch of numbers. By requiring $H^2 = P^2$ he found the Pauli matrices. He thought that he had discovered the origin of spin. He was not correct: it is not the case that *every* particle has to have spin, but he was right for particles that satisfy the Dirac equation, i.e. Fermions.

We still have $H^2 = P^2$, and we don't have a mass. So we try $H = \vec{\alpha} \cdot \vec{P} + \beta m$ for some $\beta$ such that $\beta^2 = 1$.

$$\begin{aligned}
H^2 = P^2 &+ m^2 \\
&+ P_x m(\sigma_x\beta + \beta\sigma_x) \\
&+ P_y m(\sigma_y\beta + \beta\sigma_y) \\
&+ P_z m(\sigma_z\beta + \beta\sigma_z)
\end{aligned}$$

So $\beta$ anticommutes with all the Pauli matrices: unfortunately there is no $2 \times 2$ matrix that anticommutes with all the Pauli matrices.

**Theorem 27** (No $2 \times 2$ matrix anticommutes with the Pauli matrices)**.** *We can find 3* $2 \times 2$ *non-trivial matrices that anticommute, but not 4.*

*Proof.* The Pauli matrices anticommute, so the first part of the theorem is immediate. To show there is not a $4th$ matrix, try to find $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ that anticommutes with the Pauli matrices. We start with $\sigma_1$.

$$\begin{aligned}
\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} &= - \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\
\begin{pmatrix} a & b \\ -c & -d \end{pmatrix} &= - \begin{pmatrix} a & -b \\ c & -d \end{pmatrix}, \text{ whence} \\
a =& 0 \text{ and} \\
d =& 0
\end{aligned}$$

Next we try $\sigma_2$:

$$\begin{aligned}
\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix} &= - \begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\
\begin{pmatrix} c & 0 \\ 0 & b \end{pmatrix} &= - \begin{pmatrix} b & 0 \\ 0 & c \end{pmatrix}, \text{ which reduces to:} \\
c &= - b
\end{aligned}$$

But this makes our matrix be $\begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix}$, which is a $\sigma_3$ multiplied by a scalar. The only way it can anticommute with $\sigma_3$ is if $b = 0$, which is trivial. $\square$

We have two choices: we can give up, or we can introduce another 2 valued degree of freedom, so we would use $4 \times 4$. It turns out that this will work: $4 \times 4$ *has nothing to do with space being 4 dimensional; it is just a mathematical fact about anti-commuting matrices.*

There are many different and equivalent representation of the Dirac matrices, which must satisfy:

$$\begin{aligned}
\alpha_x^2 =& 1 \\
\alpha_y^2 =& 1 \\
\alpha_z^2 =& 1 \\
\beta^2 =& 1
\end{aligned}$$

and

$$
\begin{aligned}
[\alpha_y, \alpha_z] =& 0 \\
[\alpha_z, \alpha_x] =& 0 \\
[\alpha_x, \alpha_y] =& 0 \\
[\alpha_x, \beta] =& 0 \\
[\alpha_y, \beta] =& 0 \\
[\alpha_z, \beta] =& 0
\end{aligned}
$$

Here is one solution, the Pauli representation of the Dirac matrices. This is not the representation that Dirac used.

$$
\vec{\alpha} = \begin{pmatrix} \vec{\sigma} & 0 \\ 0 & -\vec{\sigma} \end{pmatrix}
$$

$$
\alpha_x = \begin{pmatrix} \sigma_x & 0 \\ 0 & -\sigma_x \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}
$$

$$
\alpha_y = \begin{pmatrix} \sigma_y & 0 \\ 0 & -\sigma_y \end{pmatrix} = \begin{pmatrix} 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix}
$$

$$
\alpha_z = \begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}
$$

$$
\beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}
$$

We actually have a Hamiltonian whose square is $P^2 + m^2$, the relativistic law.

Questions:

- For the 1 dimensional case we had to introduce an additional degree of freedom, "spin". Actually this spin-like thing isn't really spin, which requires 3 dimensions.

- For the 3 dimensional case we have yet another additional degree of freedom. What is it? Something called "helicity". When we tried $H = \vec{\alpha} \cdot \vec{P} + \beta m$, we found that for positive energy, $\sigma$ has to be aligned with momentum. We introduced another species which has its spin reversed: this is said to be the opposite *chirality*.

**Definition 28** (Helicity)**.** *Physically, the helicity operator can be thought of as a projection operator of spin along the direction of motion* $h = \frac{\vec{S}\cdot\vec{P}}{|P|}$.

**Definition 29** (Chirality)**.** *Chirality is the relationship between the directions of spin and momentum. If the directions are the same, the chirality is right handed, otherwise left handed.[5]*

We coupled the left and right handed particles using the off diagonal entries in $\beta$, and the mass. So mass is a coupling between left and right. For massless particles, chirality is conserved.[Sort out helicity]

What is $\alpha$? Each component of $\alpha$ is a matrix, with two eigenvalues $+1$ and two $-1$. What physical quantity does it represent? They don't commute, so we can't measure them simultaneously. What is the physical meaning of a component? It is the velocity, so the expectation value in any state is the velocity of the particle.

For any operator $L$:

$$\begin{aligned}
\dot{L} =& i[H, L], \text{ for any operator L. We'll consider} \\
\dot{X} =& i[\alpha P + \beta m, X] \\
=& i\alpha[P, X] \\
=& \alpha_x
\end{aligned}$$

So it is the velocity of particle! Velocity is not the same as momentum for electron: it can only be $\pm 1$ in any direction.

$$\begin{aligned}
\frac{d}{dt}\alpha_x =& [H, \alpha_x] \\
\neq& 0 \text{ because H contains } \alpha_y \text{ and } \alpha_z
\end{aligned}$$

so $\alpha_x$ is not conserved. The electron does a spiral movement called Zitterbewegung. (But the time average turns out to be what you'd expect). And $\alpha$ is not the spin. The actual spin turns out to be $\begin{pmatrix} \vec{\sigma} & 0 \\ 0 & \vec{\sigma} \end{pmatrix}$.

Dirac discovered a multiplicity of different states of the electron, one had to do with chirality, the other to do with actual spin.

Question: doesn't the failure to conserve velocity violate Newton's law? No: Newton's law is about conservation of *momentum*.

### 10.2.3 Dirac Electron and negative energy

The Dirac equation can have both positive and negative energy, even in the 1 dimensional case, with no $\alpha$ matrix and a right moving electron. We don't need 3 dimensions for interesting effects.

$$H = P, \text{ right moving electron; what if } P < 0?$$

Negative energies are very dangerous. The ground state, the vacuum, is not necessarily the state with no particles: it is the state with lowest energy. When you suck out all the energy from space, what is left is the vacuum. Clearly the vacuum had better be the lowest possible energy, otherwise you could lower the energy past the vacuum. With negative energy we can lower the energy indefinitely! What does negative energy mean? The vacuum is supposed to be empty space, so there should be nothing to remove, no way to lower its energy further.

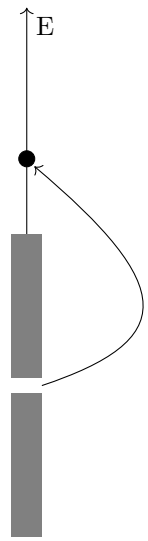
Dirac *defined* the vacuum to be the state of lowest energy. But what about all these negative energy particles? They lower energy past zero. Dirac knew that these particles were Fermions. The lowest energy would be if he filled all possible states with negative energy particles; every negative energy particles gives us negative energy, and if we fill all the negative energies states, that is as negative as it can get. Since they are Fermions, if state filled we can't keep going. Vacuum must have every negative state occupied; then there is no room to lower energy further. We can add positive energy electron, or move a negative energy electron to positive–Figure 29b. (This was an intuitive way to describe some mathematics. Modern physicists don't like this line of reasoning, but we will follow it.)

In Figure 29 we have filled all negative energy levels, which stabilizes vacuum. We can put in a positive energy electron, Figure 29a, or take out a negative energy electron and leave a hole, Figure 29b

Figure 29: TheDirac Sea: all negative energy states have been filled.

(a) The Dirac Sea, with one positive energy electron added

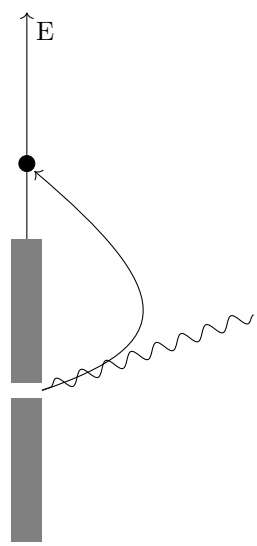
(b) Pluck out one negative energy electron from Dirac Sea, and get positive charge and positive energy hole (positron)



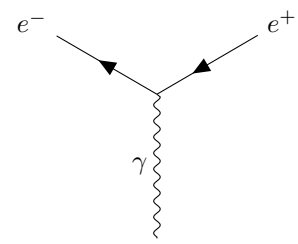

Imagine photon coming along and hitting negative energy electron in vacuum: we get an electron and a hole–Figures 30a and 30b.

Figure 30: Photon hitting -ve energy electron in vacuum

(a) Photon displaces negative energy electron and produces +ve charge and +ve energy hole (positron)




(b) Feynman diagram for (a). By convention, positron points down.




This is the Dirac theory of electrons. Let us know talk about the field operators that describe them. They are built exactly the same way as the Boson field operators.

$$\begin{aligned}\Psi &= \int_{-\infty}^{\infty} dp a^-(p) e^{-ipx} \text{Fermion operators anticommute.} \\ &= \underbrace{\int_{-\infty}^{0} dp a^-(p) e^{-ipx}}_{\text{negative energies}} + \underbrace{\int_{0}^{\infty} dp a^-(p) e^{-ipx}}_{\text{positive energies}}\end{aligned}$$

But the mathematics for Fermions doesn't distinguish creation and annihilation operators, because they anticommute. Introducing $b^+(p)$, the creation operator for positrons:

$$\Psi = \int_0^{\infty} dp b^+(p) e^{ipx} + \int_0^{\infty} dp a^-(p) e^{-ipx}$$

Consider the interaction of Figure 31a.

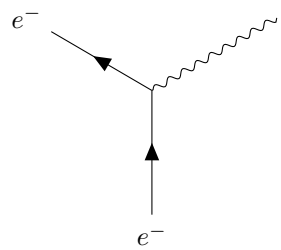
$a^+ a^- A$, which comes from a term in the Hamiltonian:

$\Psi^\dagger(x)\Psi(x)A$,

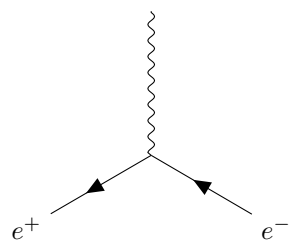
But this also describes the interactions of Figures 31a, 31b, and 31c.

Figure 31: Possible outcomes of $\Psi^\dagger(x)\Psi(x)A$: Feynman diagrams.

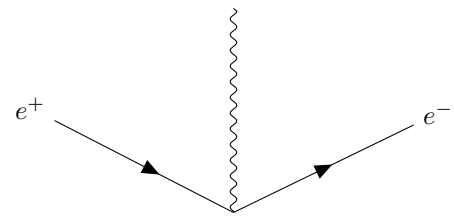
(a) Electron scattering: $a^+a^-A$




(b) Positron annihilation




(c) Electron, photon, and positron created




This was pretty much the start of Feynman diagrams (back in Dirac's day)! But Feynman figured out how to evaluate them.

How do modern physicists think of the Dirac Sea? They just forget the Dirac Sea and replace creation operators for negative energy with annihilation operators for positive. Then the theory becomes completely symmetric for electrons and positrons.

We have the same pattern in solid state physics. If you have crystal with lots of electrons, put all electrons into lowest available states–ground state. We have the Fermi Sea. Now we can take an energy out of sea and kick into higher state, leaving a hole which behaves like a particle. But Dirac did this first!

One way to look at the Dirac equation is that there are two spins and two charges (electron/positive); multiplying we get 4 components. This isn't the whole story, but it can be useful.

Question about Cosmological constant. You can think of it as energy in the Dirac Sea, or as the zero point oscillation of field. Photons aren't thought of as part of Dirac Sea, but they contribute $\frac{1}{2}\hbar\omega$ for every oscillating mode. For photons background energy is positive, for Fermions it is negative. You could think of them canceling, but Fermions have mass, photons don't, so they don't cancel. You are always left with some huge amount of energy left over.

Keep in mind that the only time zero point energy comes into anything is when that energy is gravitating. For all other purposes we can ignore it because only differences matter.

# Glossary

$\mathbb{Z}$ Set of all integers.

**Dirac Sea** All negative energy states are filled, If you want to get the lowest energy state, fill Dirac Sea with negative energy electrons..

**Fermi Sea** The electrons in the Fermi Sphere are said to be in the Fermi Sea.

**QFT** Quantum Field Theory.

**QM** Quantum Mechanics.

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